

Batch: C3 Roll No.: 16010123325**Experiment No. 5****Grade: AA / AB / BB / BC / CC / CD / DD****Title:** Experimenting with Advanced Cloud Technologies using VMware Hands-on Labs (HOL)**Objective:**

To configure and experiment with advanced cloud technologies using VMware Hands-on Labs platform, focusing on virtualization, cloud foundation architecture, and enterprise data center management.

Expected Outcome of Experiment:

CO	Outcome
CO5	Configure and experiment with advanced cloud technologies

Books/ Journals/ Websites referred:

1. VMware Hands-on Labs Portal –
<https://hol.vmware.com>
2. VMware Cloud Foundation Documentation
3. VMware vSphere Official
DocumentationVMware Learning Portal
Resources

Related Theory: -**Advanced Cloud Technologies**

- Advanced cloud technologies extend beyond basic IaaS and PaaS models to include:
- Software-Defined Data Centers (SDDC)
- Virtualization at compute, storage, and network levels
- Hyper-converged infrastructure
- Cloud orchestration and automation

- Multi-cloud integration

VMware Cloud Foundation (VCF)

VMware Cloud Foundation is an integrated software stack that combines:

- vSphere (Compute Virtualization)
- vSAN (Storage Virtualization)
- NSX (Network Virtualization)
- SDDC Manager (Automation & Lifecycle Management)

It enables:

- Private cloud deployment
- Hybrid cloud integration
- Centralized cloud operations
- Automated infrastructure provisioning

VMware vSphere

VMware vSphere is a server virtualization platform that allows:

- Creation of Virtual Machines (VMs)
- Resource pooling
- High availability (HA)
- Distributed Resource Scheduling (DRS)
- VM migration (vMotion)

Implementation Details

Students performed the experiment in two parts using VMware Hands-on Labs (HOL) platform.

Part A**Lab Performed: VMware vSphere in VMware Cloud Foundation 9.0 - Advanced Topics (HOL-2637-02-VCF-L)**

This is a hands-on lab (HOL) module that explores advanced features of VMware's integrated private cloud platform. It focuses on optimizing and extending vSphere capabilities within VMware Cloud Foundation (VCF) 9.0, a unified software-defined data center (SDDC) stack.

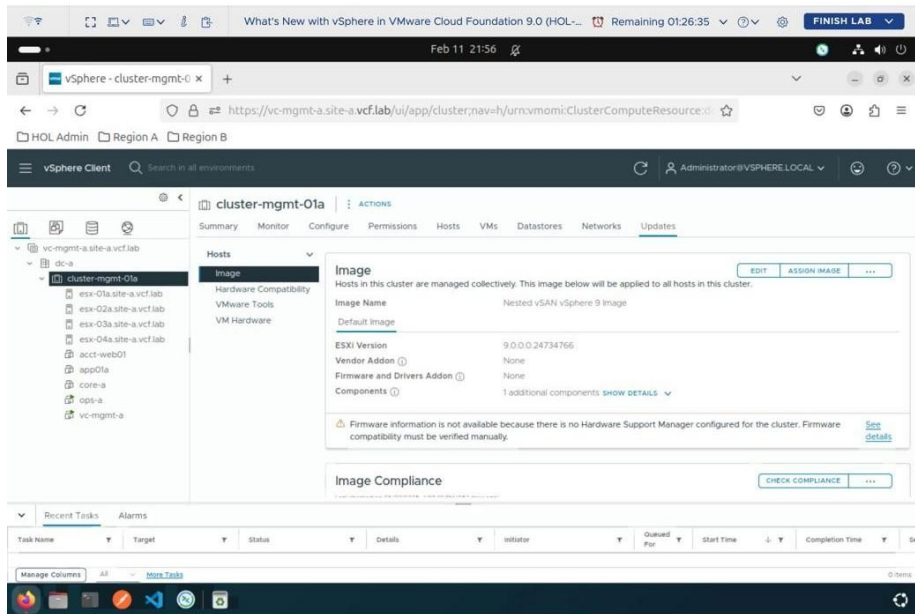
Aim of Lab

To explore advanced features of VMware vSphere within VMware Cloud Foundation 9.0 environment including:

- Cluster configuration
- Resource allocation policies
- Distributed switching
- VM migration
- High Availability configuration
- Storage policies

Steps Performed (Screenshots)-

2.1 Cluster Updates Section



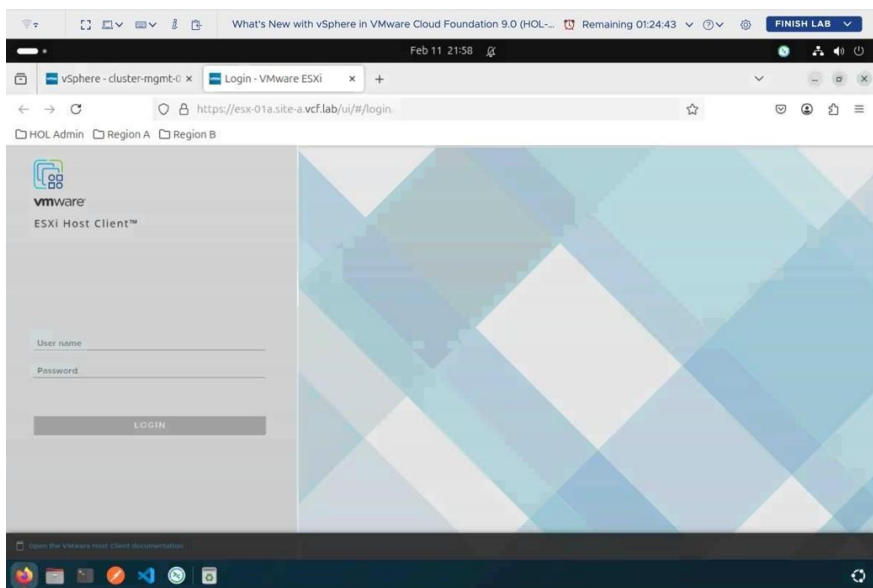
OPEN THE CLUSTER UPDATES SECTION

- Click on the cluster 'cluster-mgmt-01a'.
Note: You may need to expand the vCenter and Datacenter objects in order to see the cluster. This can be done by clicking the arrow icon to the left of each level to expand it.
- Click **Updates** in the right hand section of the cluster details.
- This is the cluster image details. Here you can see how vSphere 9 manages the hosts at a cluster level using ESX images. On this screen you can manage which image is used, add additional vendor components and check the compliance level of each host in the cluster as well as stage cluster image updates to speed up remediation.

vSphere Lifecycle Manager can stage update payloads to the hosts in advance of remediation. Staging can be performed without maintenance mode. This reduces the time needed for the hosts to spend in maintenance mode. Firmware payloads can also be staged with integration from the Lifecycle Manager.

Hi! How may I help you?

2.2 Connect to an ESX Host Client



CONNECT TO AN ESX HOST CLIENT

We will now connect to an ESX Host directly. Depending on what steps you have already completed in the lab manual, you may need to open a new browser tab. You can do this by clicking the "+" symbol on the Firefox browser.

- If you need to open a new browser tab, you can click the "+" symbol to do so. Otherwise you can move to the next step.

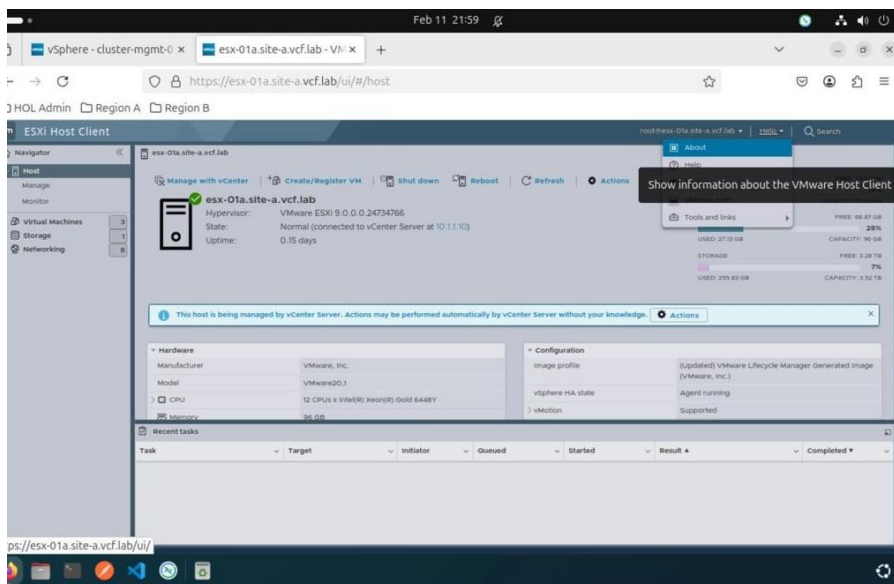
As a reminder, you can select text in this manual and drag them into the console to reduce the need to type information. That can be helpful for entering the URL in step 2.

- Enter the URL for the host. The URL is: `https://esx-01a.site-a.vcf.lab`. Press **ENTER** on your keyboard after entering the URL.

This should launch the host management UI for the host.

Hi! How may I help you?

2.3 Information about VMware Host Client



The screenshot shows the VMware Host Client interface. The 'About' menu is open, displaying system information such as Hypervisor (VMware ESXi 9.0), State (Normal), and Uptime (0.15 days). The 'Tools and links' section includes links for Help, Update, VMware.com, and Tools and links. The 'Recent tasks' table is also visible.

OPEN THE ABOUT MENU

- Click on 'Help' in the top right corner of the screen.
- Click on 'About' in the menu.

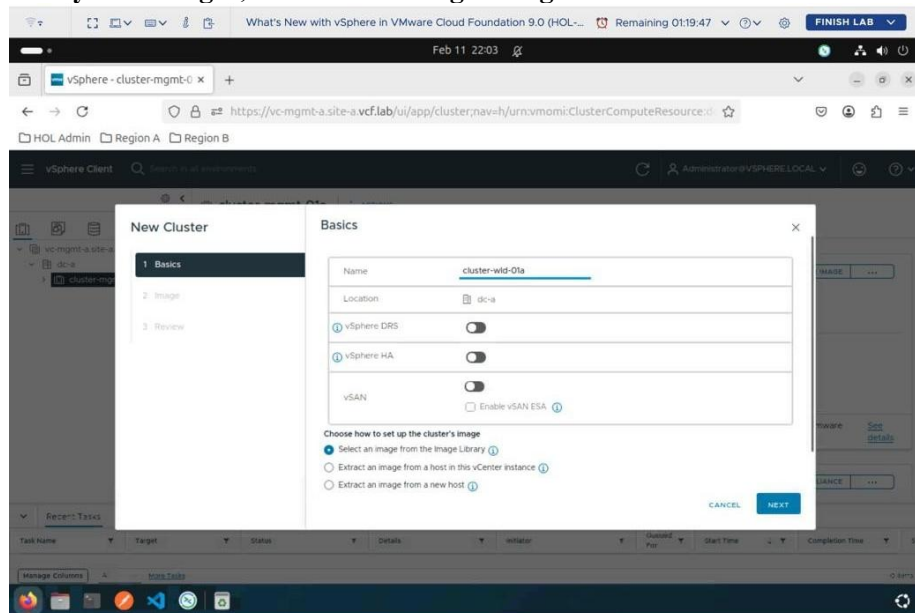
For this section we will explore some of the new UI visual changes. These settings are accessed via the About menu in the UI. Let's go there now so we can see these new enhancements.

1. Click on 'Help' in the top right corner of the screen.
2. Click on 'About' in the menu.

Hi! How may I help you?

2.4 New Cluster

We utilize the same Cluster Image already created for the initial cluster, this a key part of Lifecycle manager, we can use a single image to ensure al



The screenshot shows the VMware vSphere Client interface with the 'New Cluster' wizard open. The 'Basics' tab is selected, showing fields for Name (cluster-wid-01a), Location (dc-a), and vSphere HA settings. The 'Choose how to set up the cluster's image' section is also visible.

NEW CLUSTER - BASICS

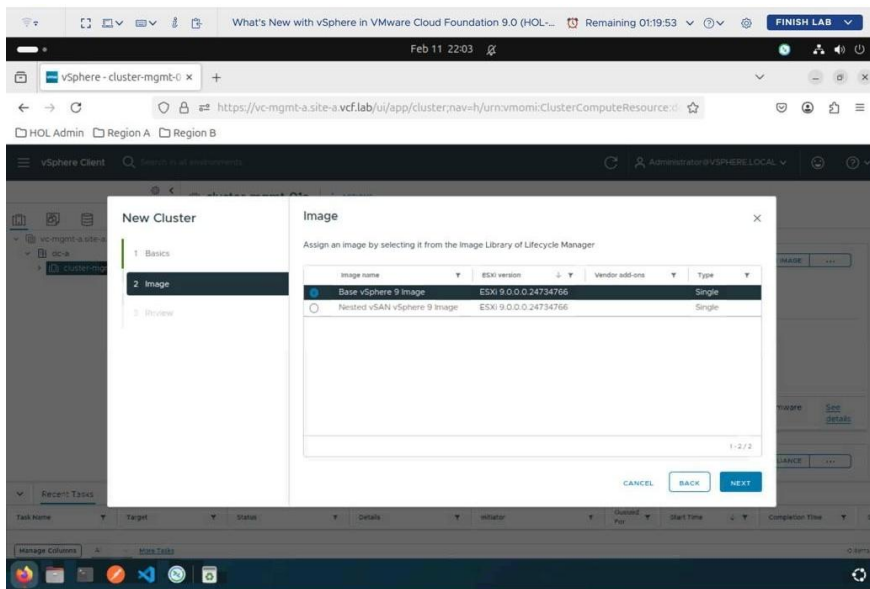
- Name the new cluster 'cluster-wid-01a'.
- Ensure the 'Select an image from the Image Library' option is selected.
- Click 'NEXT'.

In this new cluster, we will utilize the same Cluster Image already created for the initial cluster. This is a key part of Lifecycle manager; we can leverage a single image, based on a host in our environment, to ensure that all existing and new hosts are consistent. This image also includes any vendor specific components such as those provided by Dell, HPC, etc. When we created the 'cluster-mgmt-01a' cluster, we had vLCM capture in image from the 'esx-01a' host. We will now leverage this same image for our new cluster.

1. Name the new cluster 'cluster-wid-01a'.
2. Ensure the 'Select an image from the Image Library' option is selected.
3. Click 'NEXT'.

Hi! How may I help you?

Existing and new hosts are consistent



NEW CLUSTER - IMAGE

Assign an image by selecting it from the Image Library of Lifecycle Manager

Image name	ESXi version	Vendor add-ons	Type
Base vSphere 9 Image	ESXi 9.0.0.0.24734766		Single
Nested vSAN vSphere 9 Image	ESXi 9.0.0.0.24734766		Single

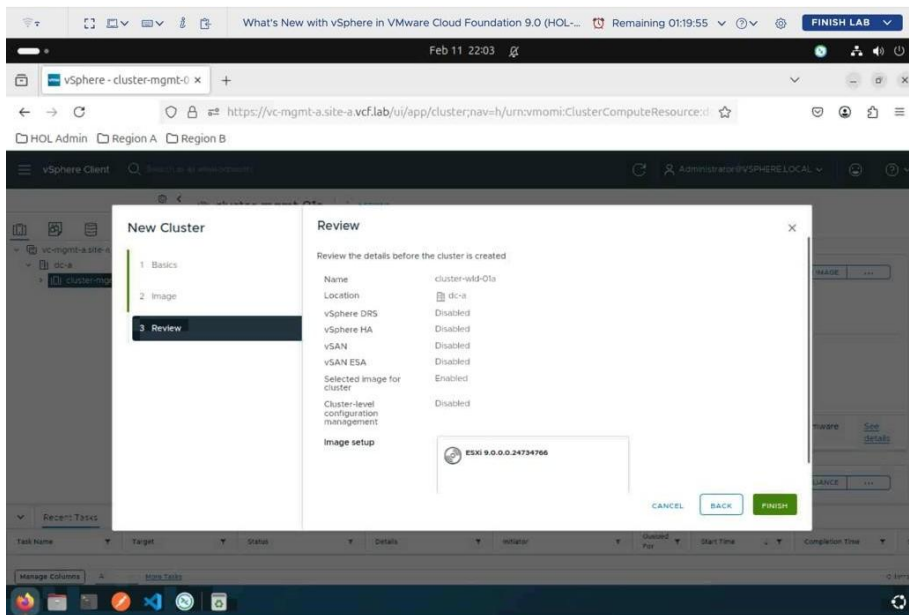
1-2/2

NEW CLUSTER - IMAGE

This is the key step in the new cluster creation process, here we will select the image we want this cluster to use for all the hosts in this cluster. As mentioned previously, we have already prepared an image that includes all the key components we will need.

1. Select the 'Base vSphere 9 Image' image.
2. Click 'NEXT'.

Hi! How may I help you?



NEW CLUSTER - REVIEW

Review the details before the cluster is created

Name	cluster-oid-01a
Location	dc-a
vSphere DRS	Disabled
vSphere HA	Disabled
vSAN	Disabled
vSAN ESA	Disabled
Selected image for cluster	Enabled
Cluster-level configuration management	Disabled

Image setup

ESXi 9.0.0.0.24734766

NEW CLUSTER - REVIEW

This is our final review of our new cluster creation, here we see a summary of the options we select, including the image that will be used for the cluster. With this image selected, we will be able to remediate the entire cluster using parallel remediation.

1. Ensure we have an image attached to the cluster.
2. Click 'FINISH'.

Hi! How may I help you?

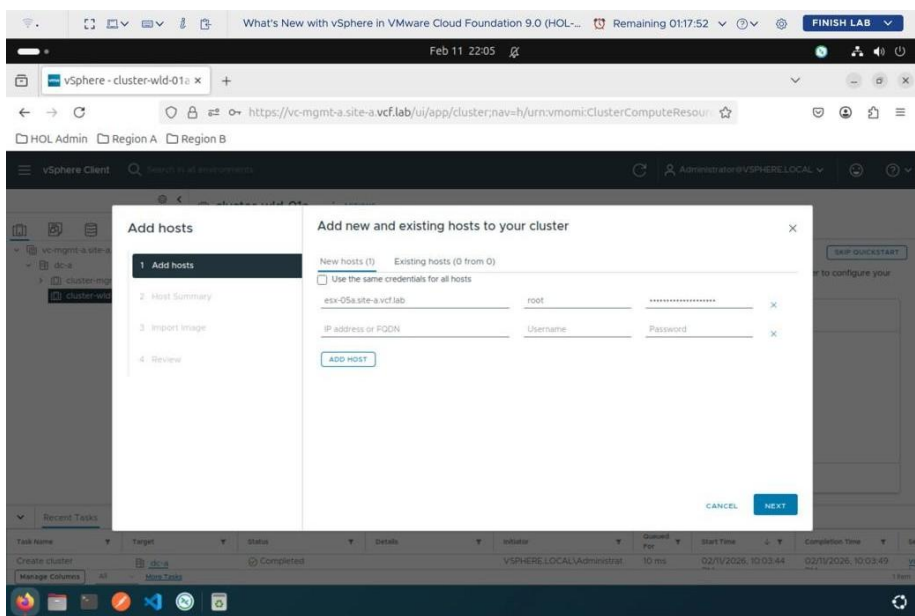


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(Somaiya Vidyavihar University)
Department of Computer Engineering



2.5 Add Second Host to the Cluster

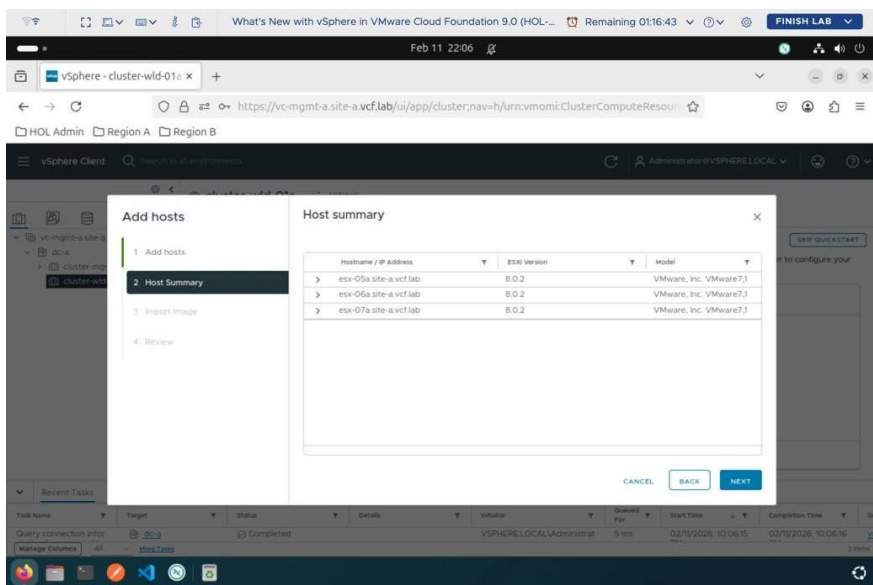
Adding hosts with same credentials



ADD A SECOND HOST TO THE CLUSTER

- We can save ourselves some typing as all of our hosts use the same local credentials to login by checking the **Use the same credentials for all hosts** option.
- Populate the hostname for the remaining hosts (be sure you clicked the 'ADD HOST' button enough time for all the hosts)
 - esx-06a.site-a.vcf.lab
 - esx-07a.site-a.vcf.lab
- Click **NEXT**

Hi! How may I help you?

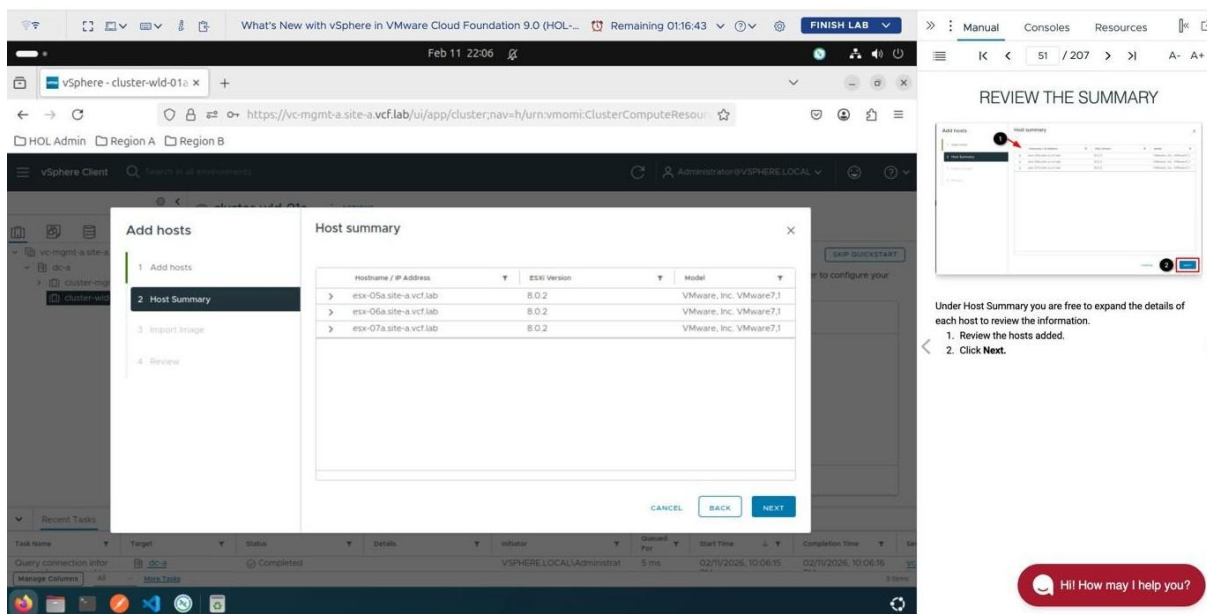


REVIEW THE SUMMARY

Under Host Summary you are free to expand the details of each host to review the information.

- Review the hosts added.
- Click **Next**.

Hi! How may I help you?



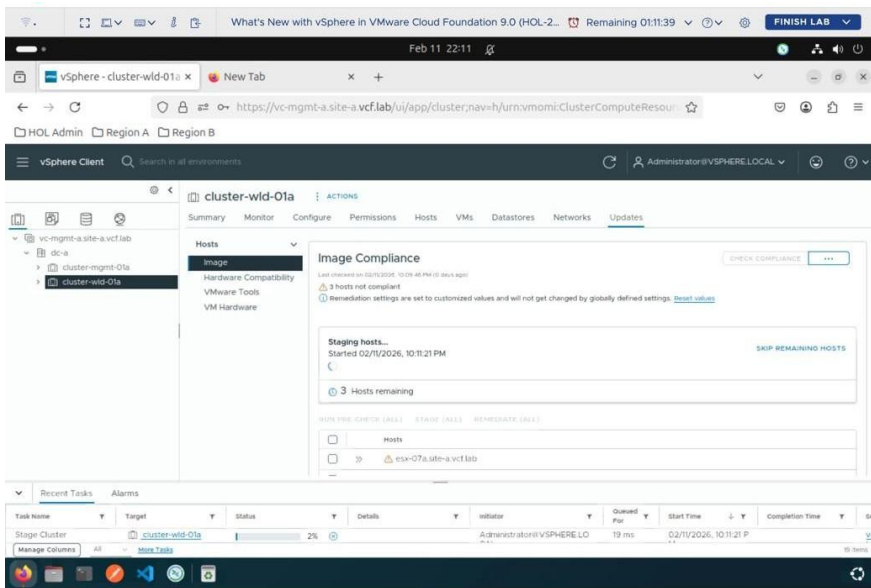
The screenshot shows the vSphere Client interface with the 'Add hosts' wizard. The 'Host summary' step is active, displaying a table of hosts. A 'REVIEW THE SUMMARY' sidebar is visible on the right.

Hostname / IP Address	ESXi Version	Model
esx-05a.site-a.vcf.lab	8.0.2	VMware, Inc. VMware7.1
esx-06a.site-a.vcf.lab	8.0.2	VMware, Inc. VMware7.1
esx-07a.site-a.vcf.lab	8.0.2	VMware, Inc. VMware7.1

Under Host Summary you are free to expand the details of each host to review the information.

1. Review the hosts added.
2. Click Next.

2.6 Stage Hosts & Remediate them



STAGE HOSTS AND REMEDIATE THEM

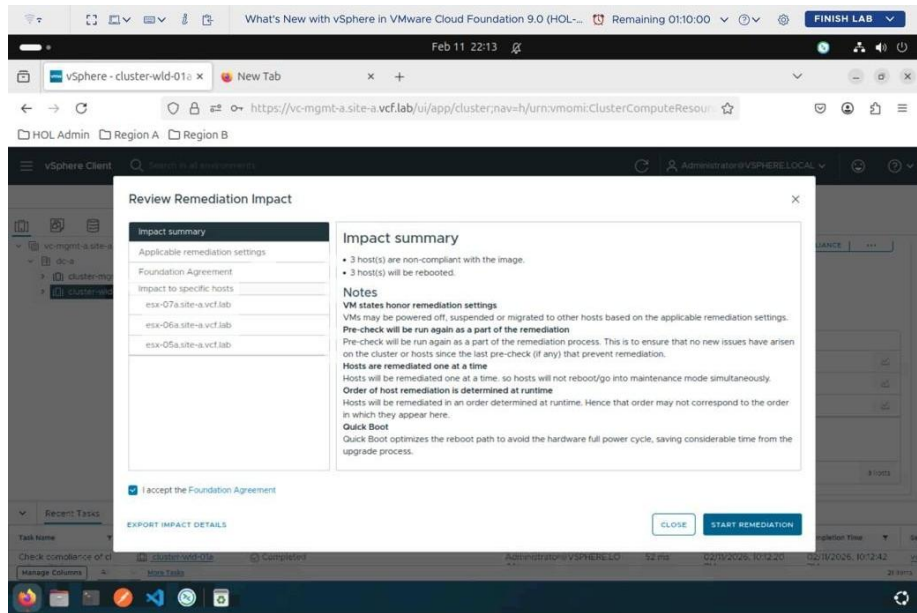
Let us stage the hosts and then remediate.

1. Click **STAGE ALL**
 - a. Observe all hosts show as **STAGED** (This step may take several minutes, if you would like to do so, you are free to move to another module while this process completes and return to this step to check that the hosts are staged)
2. Observe that the hosts that were added to cluster in the previous steps all have a green check mark to the right of them under the Image Compliance section. This indicates they are staged and ready to be remediated.
3. Click **REMEDIATE ALL**

Hi! How may I help you?

2.7 Remediation Impact

Impact Summary



The screenshot shows the vSphere Client interface with a 'Review Remediation Impact' dialog box open. The dialog box contains an 'Impact summary' section with the following details:

- Applicable remediation settings:** Foundation Agreement, Impact to specific hosts, esx-07a.site-a.vcf.lab, esx-06a.site-a.vcf.lab, esx-05a.site-a.vcf.lab.
- Impact summary:** 3 host(s) are non-compliant with the image. 3 host(s) will be rebooted.
- Notes:** VM states honor remediation settings. VMs may be powered off, suspended or migrated to other hosts based on the applicable remediation settings. Pre-check will be run again as a part of the remediation process. Pre-check will be run again as a part of the remediation process. This is to ensure that no new issues have arisen on the cluster or hosts since the last pre-check (if any) that prevent remediation. Hosts are remediated one at a time. Hosts will be remediated one at a time, so hosts will not reboot/go into maintenance mode simultaneously. Order of host remediation is determined at runtime. Hosts will be remediated in an order determined at runtime. Hence that order may not correspond to the order in which they appear here. Quick Boot: Quick Boot optimizes the reboot path to avoid the hardware full power cycle, saving considerable time from the upgrade process.

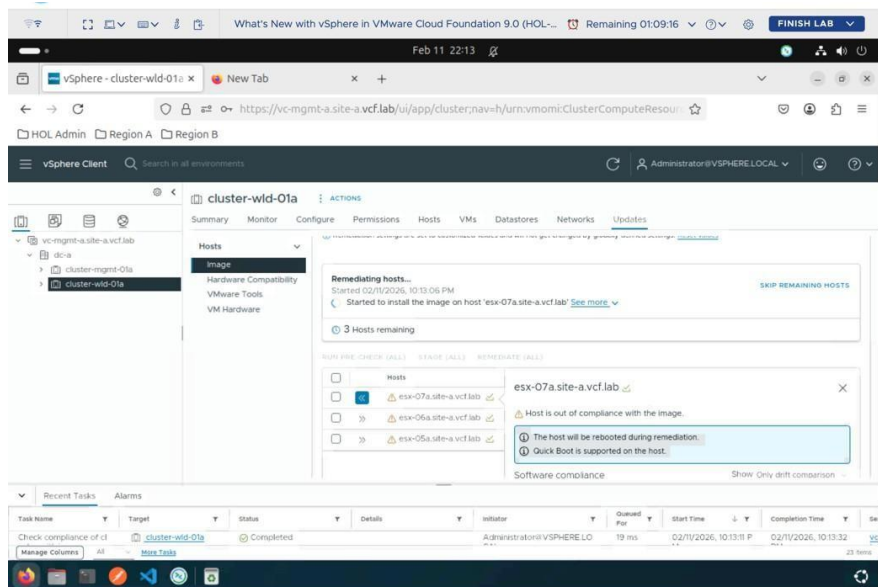
At the bottom of the dialog box, there is a checkbox for 'I accept the Foundation Agreement' and buttons for 'CLOSE' and 'START REMEDIATION'.

On the right side of the screenshot, there is a 'REMEDATION IMPACT' sidebar with a summary of the process and a list of steps:

1. Review the Impact Summary
2. Click on START REMEDIATION

A red button at the bottom right says 'Hi! How may I help you?'.

Parallel Remediation



The screenshot shows the vSphere Client interface with the 'cluster-wid-01a' cluster selected. The 'Hosts' tab is active, showing a list of hosts and their status. The 'Remediating hosts...' section indicates that the remediation process has started on host 'esx-07a.site-a.vcf.lab' and that 3 hosts remain.

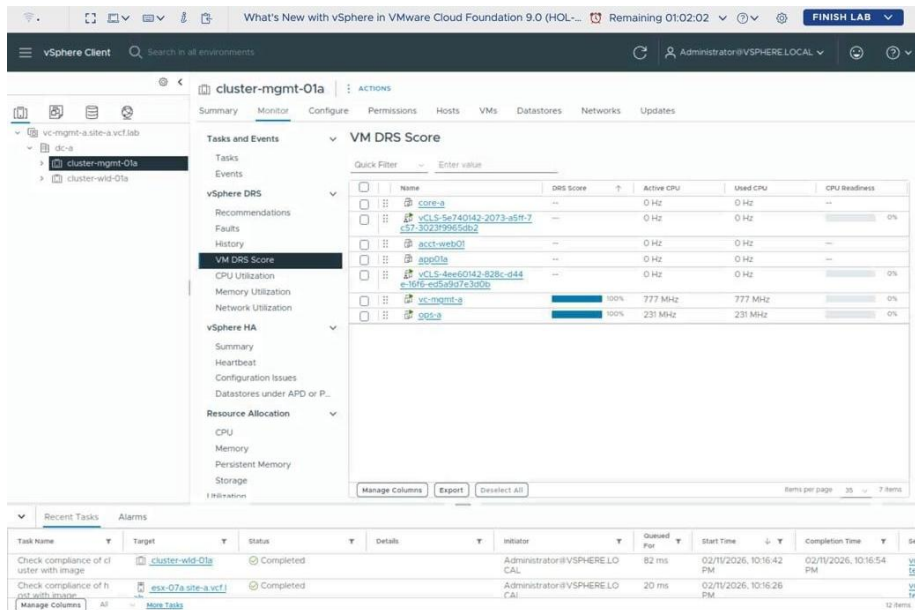
On the right side of the screenshot, there is a 'OBSERVE REMEDIATING HOSTS' sidebar with the following information:

- Image Compliance:** 3 hosts not compliant. Remediation settings are set to customized values and will not get changed by globally defined settings.
- Remediating hosts:** esx-07a.site-a.vcf.lab (10:13:32 PM).

Below the sidebar, there is a text box that reads: 'Remediation Begins - Parallel Remediation of hosts in maintenance mode. Observe that the hosts are remediating in parallel. Congratulations! You just performed parallel remediation of hosts in vSphere 9. This is the final stage of the vSphere Life Cycle Manager Parallel Remediation. Feel free to wait for the remediation to complete or move on to the next module.'

A red button at the bottom right says 'Hi! How may I help you?'.

2.7 Review DRS scores for all VMs in the cluster



Name	DRS Score	Active CPU	Used CPU	CPU Readiness
core-a	---	0 Hz	0 Hz	---
vc15-5e740142-2073-a5ff-7	---	0 Hz	0 Hz	---
cs7-30239965db2	---	0 Hz	0 Hz	---
acct-web01	---	0 Hz	0 Hz	---
app01a	---	0 Hz	0 Hz	---
vc15-4ee60142-828c-d44	---	0 Hz	0 Hz	---
vc15-ee56907e-3508	---	0 Hz	0 Hz	---
vc-mgmt-a	100%	777 MHz	777 MHz	0%
app-a	100%	231 MHz	231 MHz	0%

Recent Tasks

Task Name	Target	Status	Details	Initiator	Queued For	Start Time	Completion Time
Check compliance of cluster with image	cluster-wild-01a	Completed		Administrator@VSPHERE.LOCAL	62 ms	02/11/2026, 10:16:42 PM	02/11/2026, 10:16:54 PM
Check compliance of host with image	esx-07a.site-a.vcf	Completed		Administrator@VSPHERE.LOCAL	20 ms	02/11/2026, 10:16:26 PM	

Manual Consoles Resources

1. Under Inventory - Expand the vc-mgmt-a.site-a.vcf lab -> dc-a
2. Click cluster-mgmt-01a
3. Click Monitor
4. Click VM DRS Score, under vSphere DRS. You may need to scroll down to see this option.

Review the DRS Score for all Virtual Machines on the Cluster

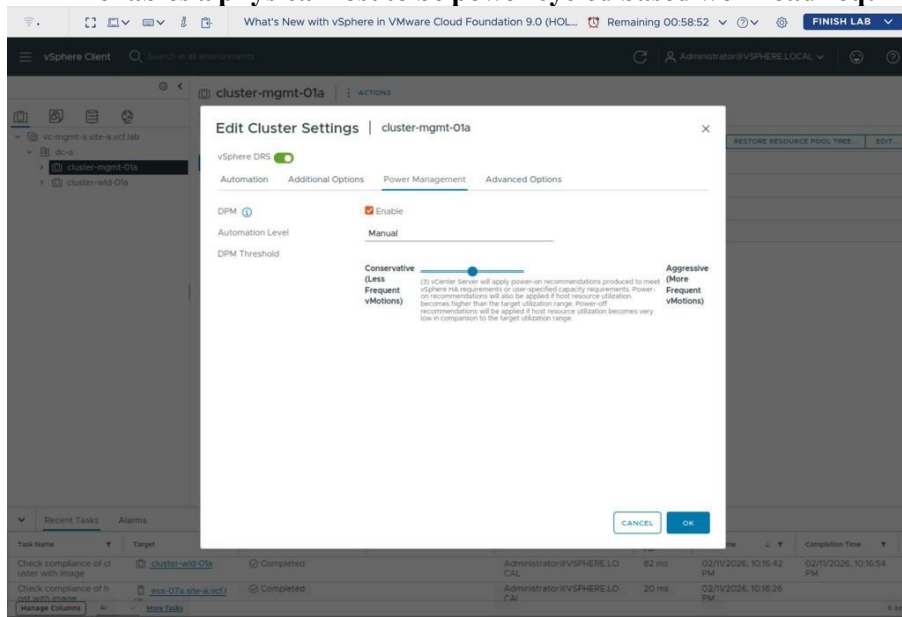
TIP! Score below 100% indicates that there is a possible improvement for that Virtual Machine, this will correct over time if DRS is working. If these values remain below 100% for certain VMs then that could indicate that those workloads are not receiving the required resources to effectively balance across the cluster

If you have knowledge about VMware DRS and the different features and functions available please continue to the next Module, if you would like to find out more please continue

Hi! How may I help you?

2.8 VSPHERE DRS – Power Management

DPM enables a physical host to be power cycled based workload requirements.



Edit Cluster Settings | cluster-mgmt-01a

vSphere DRS ☒

Automation Additional Options **Power Management** Advanced Options

DPM ☒ Enable

Automation Level Manual

DPM Threshold

Conservative (Less Frequent vMotion) Aggressive (More Frequent vMotion)

(D) vCenter Server will apply power-on recommendations produced to meet configured requirements or user specified capacity requirements. Power-on recommendations will also be applied if host resource utilization becomes higher than the target utilization range. Power-off recommendations will be applied if host resource utilization becomes very low in comparison to the target utilization range.

Manual Consoles Resources

VSPHERE DRS - POWER MANAGEMENT

1. Click Power Management
2. Check the Enable box

DPM
Distributed Power Management enables a physical host to be power cycled based on workload requirements, during peak usage physical host will be powered on based on the thresholds set by the DPM Threshold. To function, DPM requires IPMI / ILO or Wake-on-Lan to be able to power on the host. During off-peak the workloads will be consolidated onto as few hosts as possible that don't exceed the thresholds configured.

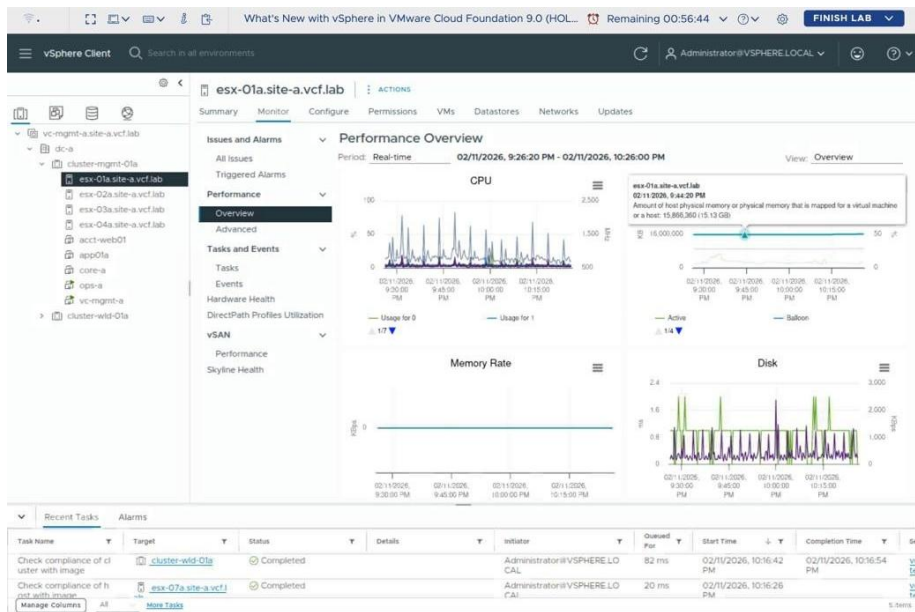
You can also create scheduled tasks to enable and deactivate DPM for a cluster using the

Schedule Task: Change Cluster Power Settings wizard.
Automation Level

Hi! How may I help you?

• **OFF** - The feature is deactivated and no

2.9 DRS Performance Overview - VMMR



The screenshot shows the vSphere Client interface with the Performance Overview for the host 'esx-01a.site-a.vcf.lab'. The left sidebar shows the navigation tree with 'esx-01a.site-a.vcf.lab' selected. The main panel displays various performance metrics: CPU (Usage for 0 and 1), Memory Rate, and Disk (Active and Subson). A table at the bottom lists recent tasks, including 'Check compliance of host with image' and 'Check compliance of host with image'.

ENHANCED DRS PERFORMANCE - VSPHERE MEMORY MONITORING (VMMR) AND REMEDIATION

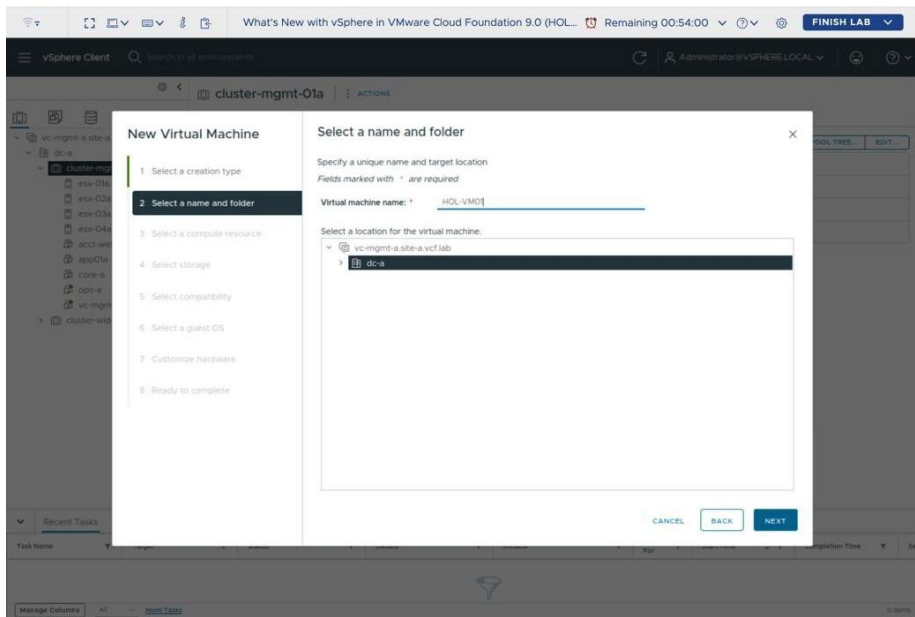
- Under Inventory - Expand **vc-mgmt-a.site-a.vcf.lab** -> **dc-a** -> **cluster-mgmt-01a**
- Click **esx-01a.site-a.vcf.lab**
- Click **Monitor**
- Click **Performance** > **Overview**
- In View: click the pull the drop-down menu
- Select **Memory**

VMMR helps bridge the need for monitoring by providing running statistics at both the VM (bandwidth) and Host levels (bandwidth, miss-rates). VMMR also provides default alerts and the ability to configure custom alerts based on the workloads that run on VMs.

VMMR collects data and provides visibility of performance statistics so you can determine if your application workload is regressed due to Memory Mode.

Hi! How may I help you?

2.10 Create new VM



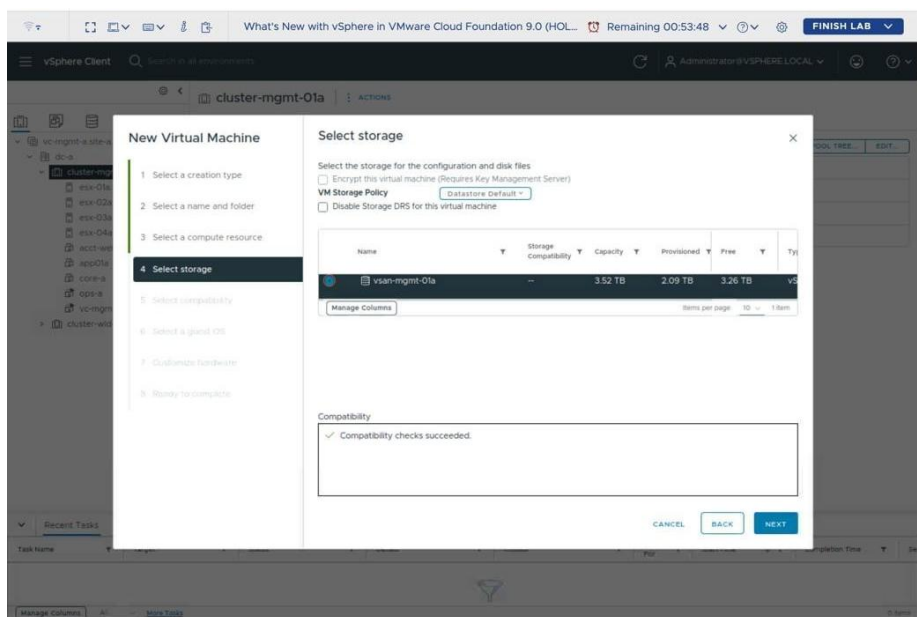
The screenshot shows the vSphere Client interface with the 'New Virtual Machine' wizard open. The wizard is at step 2: 'Select a name and folder'. The 'Virtual machine name' field is set to 'HOL-VM01'. The 'Select a location for the virtual machine' dropdown shows 'dc-a' selected. The wizard has buttons for 'CANCEL', 'BACK', and 'NEXT'.

CREATE VIRTUAL MACHINE (VM NAME)

- Replace the Virtual machine name to **HOL-VM01**
- Ensure that the Cluster **dc-a** is selected as the location.
- Click **NEXT**

Hi! How may I help you?

Select Storage

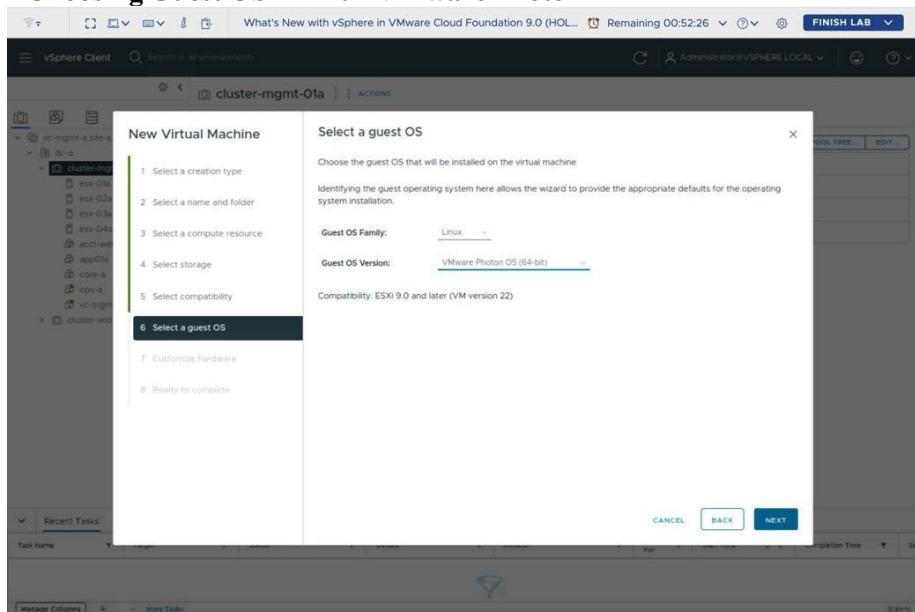


CREATE VIRTUAL MACHINE (SELECT STORAGE)

1. Select **vsan-mgmt-01a** from the storage options and leave all other options default
2. Take note that vCenter will inform you if there are compatibility issues between the VM Storage Policy and the datastore you have selected. There are other labs that discuss the storage policies in detail and I encourage you to take them to learn more about this feature. For the purposes of our lab, we are using the default storage policy and there is only one datastore that we have already insured is compatible.
3. Click **NEXT**

Hi! How may I help you?

Choosing Guest OS – Linux VMware Photon



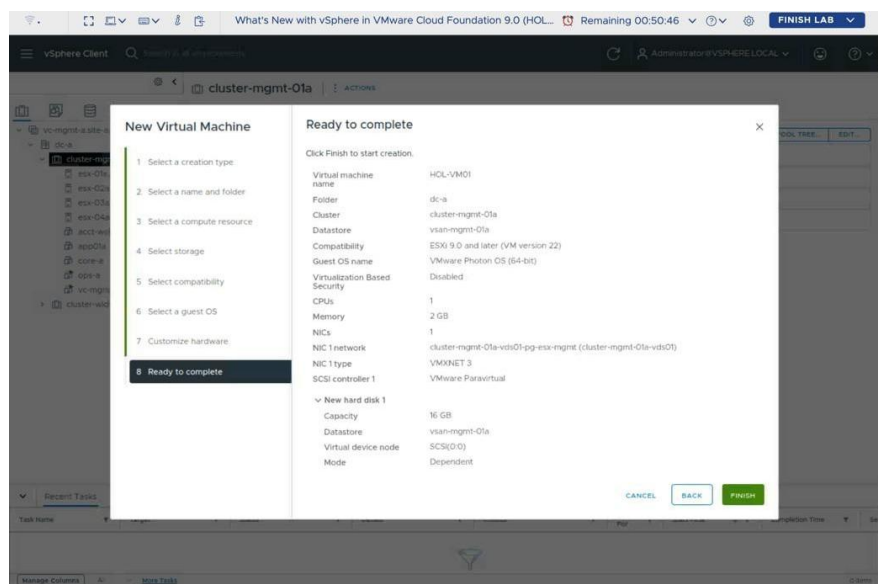
CREATE VIRTUAL MACHINE (OS)

Our virtual machine will be using VMware's Photon OS, so we need to ensure we have selected the correct OS family and version. This is important for many reasons such as ensuring the correct OS is represented in vCenter, also, this helps our wizard provide you with the correct defaults from the hardware and OS installation.

1. Click on the **Guest OS Family** dropdown and select **Linux**
2. Click on the **Guest OS Version** dropdown and select **VMware Photon OS (64-bit)**
3. Click **NEXT**

Hi! How may I help you?

CPU Topolgy

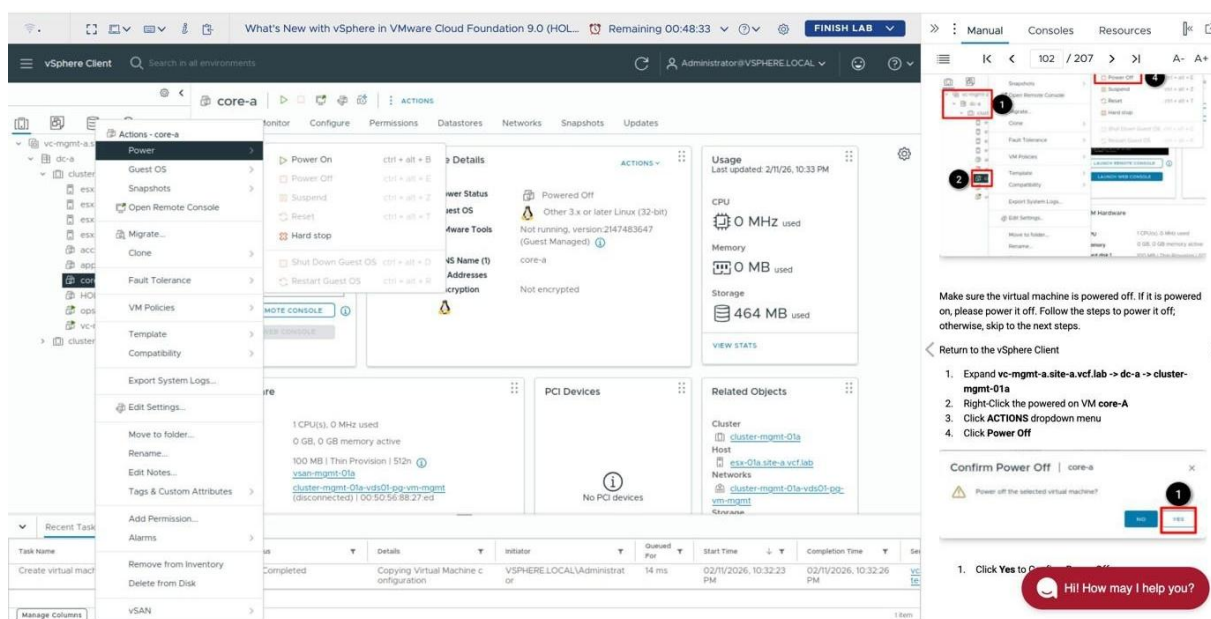


CREATE VIRTUAL MACHINE (VM OPTIONS - CPU TOPOLOGY)

1. Click on the **CPU Topology** drop down to expose the available options.
2. To change the default behavior, select the dropdown for **Cores per Socket** (We will keep the default), but you are free to review the available options. If this VM had multiple CPUs assigned, you could dictate how many cores would be assigned per socket. Since our VM only has 1 CPU the dropdown will only have 1 option.
3. Click **NEXT**

Hi! How may I help you?

2.11 Power off the VM



The screenshot shows the vSphere Client interface. The 'Power' menu is open, and the 'Power Off' option is highlighted. A confirmation dialog box is shown with the 'Yes' button highlighted.

Power Off Confirmation Dialog:

Power off the selected virtual machine?

Yes No

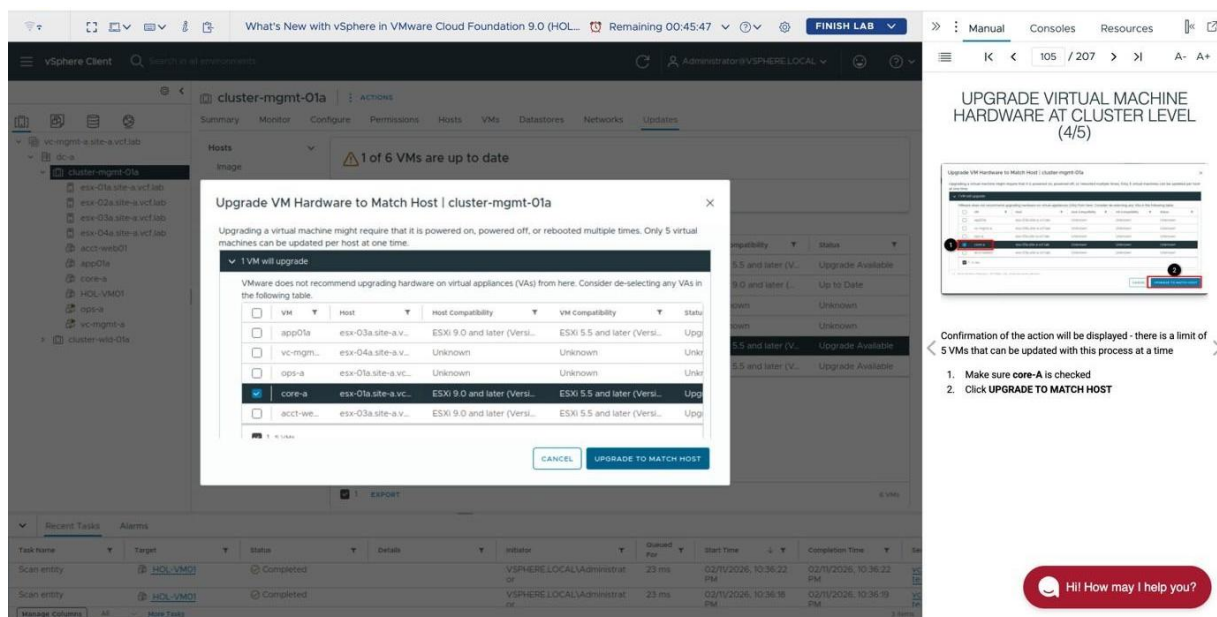
Instructions:

- Expand `vc-mgmt-a.site-a.vcf.lab` -> `dc-a` -> `cluster-mgmt-01a`
- Right-Click the powered on VM `core-a`
- Click **ACTIONS** dropdown menu
- Click **Power Off**

Confirm Power Off | `core-a`

1. Click Yes to Power Off the VM

2.12 Upgrading the VM Hardware at Cluster Level to match Host



The screenshot shows the vSphere Client interface with a modal dialog titled "Upgrade VM Hardware to Match Host | cluster-mgmt-01a". The dialog lists VMs and their compatibility with the host. The "core-a" VM is selected for upgrade.

VM	Host	Host Compatibility	VM Compatibility	Status
app01a	esx-Q3a.site-a.v...	ESXi 9.0 and later (Versi...	ESXi 5.5 and later (Versi...	Upgr...
vc-mgmt...	esx-Q4a.site-a.v...	Unknown	Unknown	Unk...
ops-a	esx-Q1a.site-a.vc...	Unknown	Unknown	Unk...
core-a	esx-Q1a.site-a.vc...	ESXi 9.0 and later (Versi...	ESXi 5.5 and later (Versi...	Upgr...
acct-we...	esx-Q3a.site-a.v...	ESXi 9.0 and later (Versi...	ESXi 5.5 and later (Versi...	Upgr...

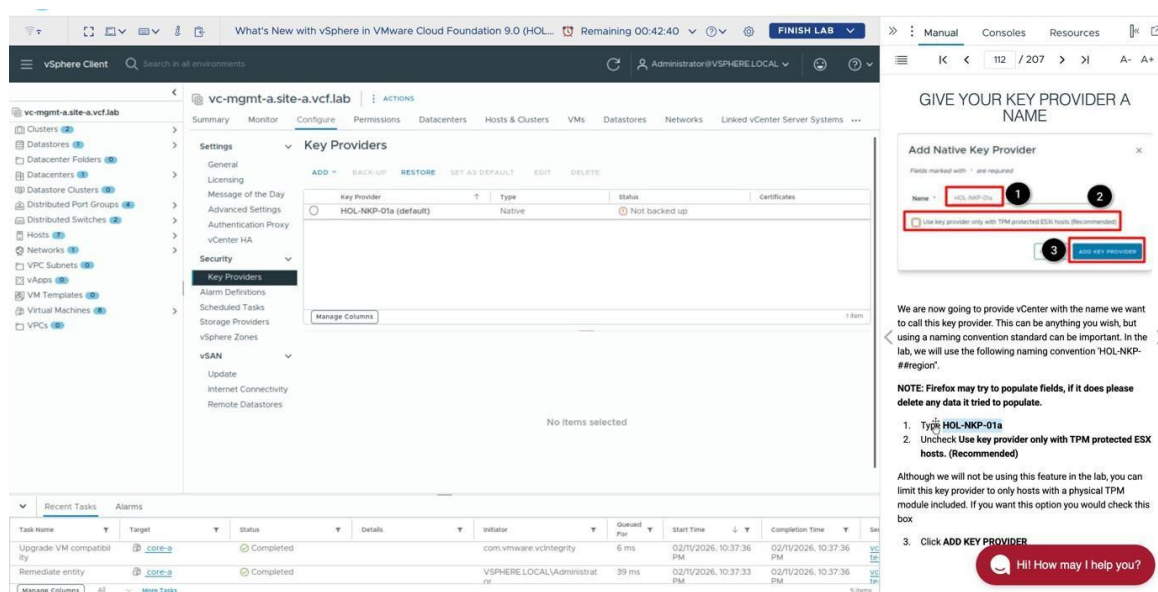
Buttons: CANCEL, UPGRADE TO MATCH HOST

Confirmation of the action will be displayed - there is a limit of 5 VMs that can be updated with this process at a time

1. Make sure **core-A** is checked
2. Click **UPGRADE TO MATCH HOST**

Hi! How may I help you?

2.13 vCenter – Providing name to key provider



The screenshot shows the vSphere Client interface with the "Key Providers" section. A modal dialog titled "GIVE YOUR KEY PROVIDER A NAME" is open, showing the "Add Native Key Provider" form.

Fields marked with * are required:

- Name: HOL-NKP-01a (1)
- Use key provider only with TPM protected ESX hosts (Recommended) (2)
- ADD KEY PROVIDER (3)

We are now going to provide vCenter with the name we want to call this key provider. This can be anything you wish, but using a naming convention standard can be important. In the lab, we will use the following naming convention HOL-NKP-#region.

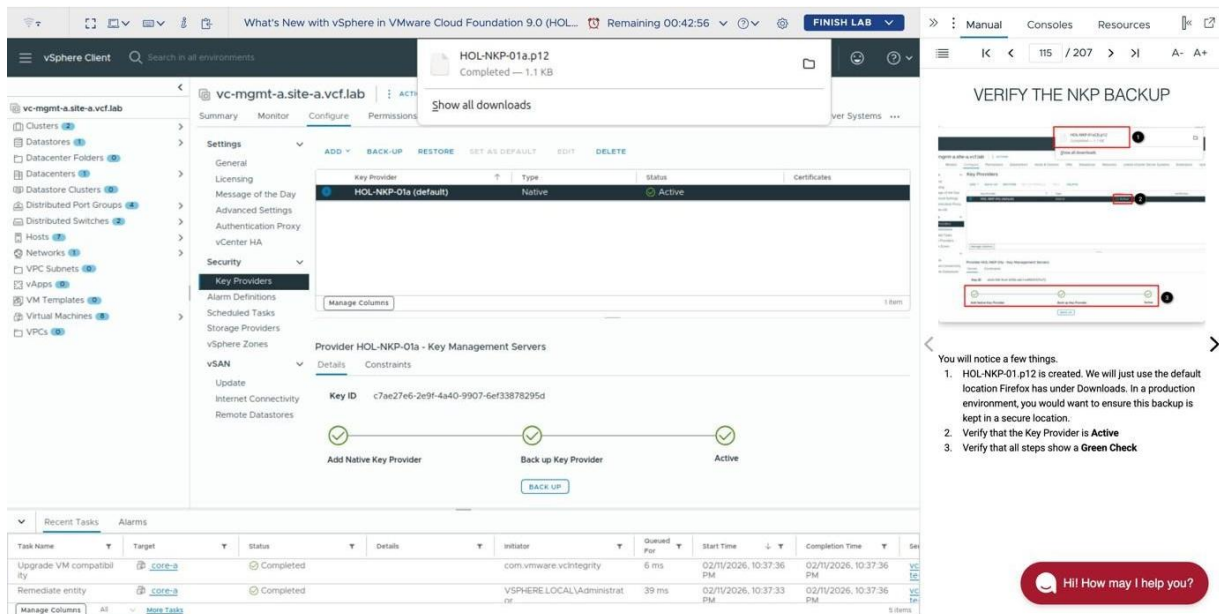
NOTE: Firefox may try to populate fields, if it does please delete any data it tried to populate.

1. Type **HOL-NKP-01a**
2. Uncheck **Use key provider only with TPM protected ESX hosts. (Recommended)**
3. Click **ADD KEY PROVIDER**

Although we will not be using this feature in the lab, you can limit this key provider to only hosts with a physical TPM module included. If you want this option you would check this box

Hi! How may I help you?

2.14 Verify the NKP Backup



The screenshot shows the vSphere Client interface with the 'HOL-NKP-01a.p12' backup file selected. The 'Verify the NKP Backup' section on the right provides a checklist of steps to ensure the backup is valid and secure.

VERIFY THE NKP BACKUP

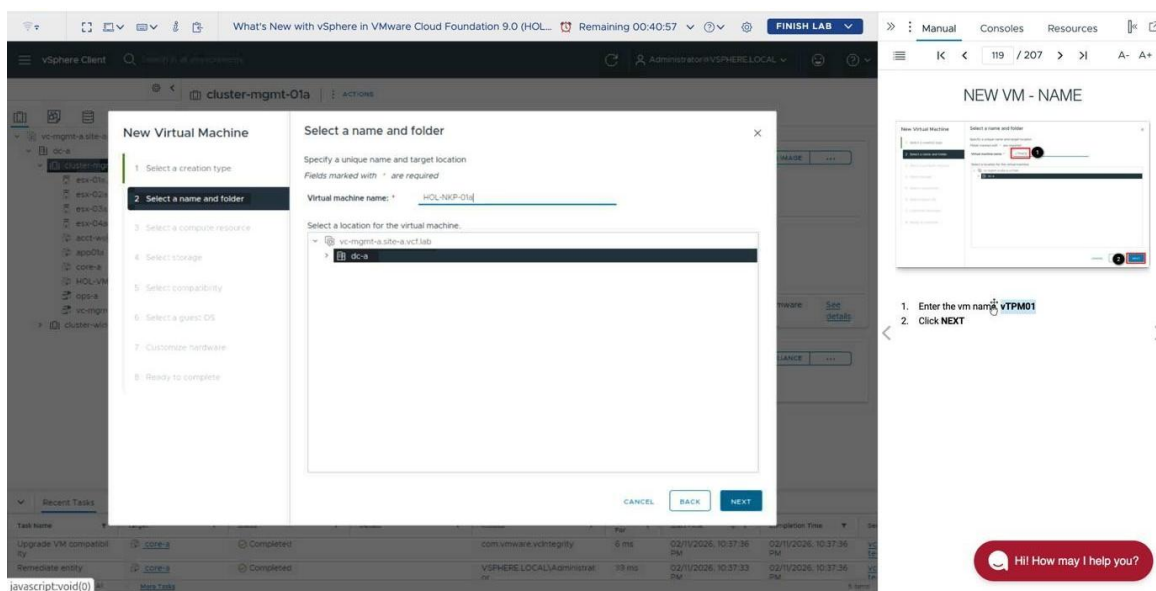
You will notice a few things.

1. HOL-NKP-01.p12 is created. We will just use the default location Firefox has under Downloads. In a production environment, you would want to ensure this backup is kept in a secure location.
2. Verify that the Key Provider is **Active**.
3. Verify that all steps show a **Green Check**.

Recent Tasks:

Task Name	Target	Status	Details	Initiator	Quarantined For	Start Time	Completion Time	Size
Upgrade VM compatibility	core-a	Completed		com.vmware.vchintegrity	6 ms	02/11/2026, 10:37:36 PM	02/11/2026, 10:37:36 PM	5.0 MB
Remediate entity	core-a	Completed		VSPHERE.LOCAL\Administrator	39 ms	02/11/2026, 10:37:33 PM	02/11/2026, 10:37:36 PM	5.0 MB

2.15 Create new VM



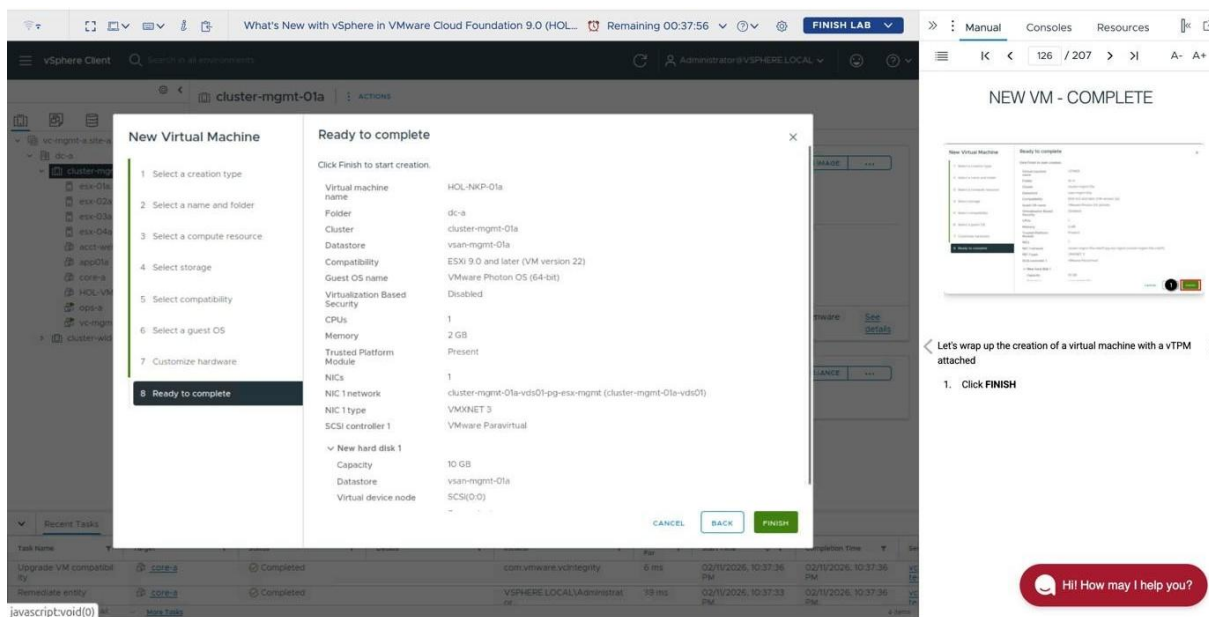
The screenshot shows the 'New Virtual Machine' wizard in the vSphere Client. The 'Select a name and folder' step is highlighted, showing the 'Virtual machine name' field with the value 'HOL-NKP-01a' and the 'Select a location for the virtual machine' dropdown set to 'dc-a'.

NEW VM - NAME

1. Enter the vm name **VTPM01**
2. Click **NEXT**

Recent Tasks:

Task Name	Target	Status	Details	Initiator	Quarantined For	Start Time	Completion Time	Size
Upgrade VM compatibility	core-a	Completed		com.vmware.vchintegrity	6 ms	02/11/2026, 10:37:36 PM	02/11/2026, 10:37:36 PM	5.0 MB
Remediate entity	core-a	Completed		VSPHERE.LOCAL\Administrator	39 ms	02/11/2026, 10:37:33 PM	02/11/2026, 10:37:36 PM	5.0 MB

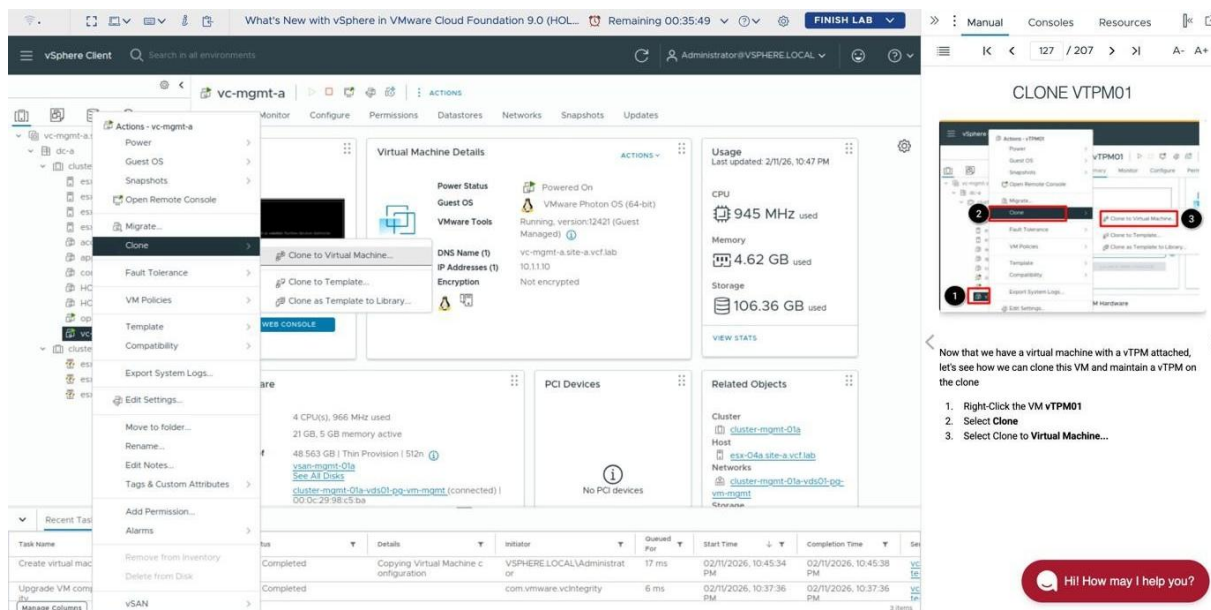


NEW VM - COMPLETE

Let's wrap up the creation of a virtual machine with a vTPM attached

1. Click **FINISH**

2.16 Cloning the VTPM01 VM created in the earlier step



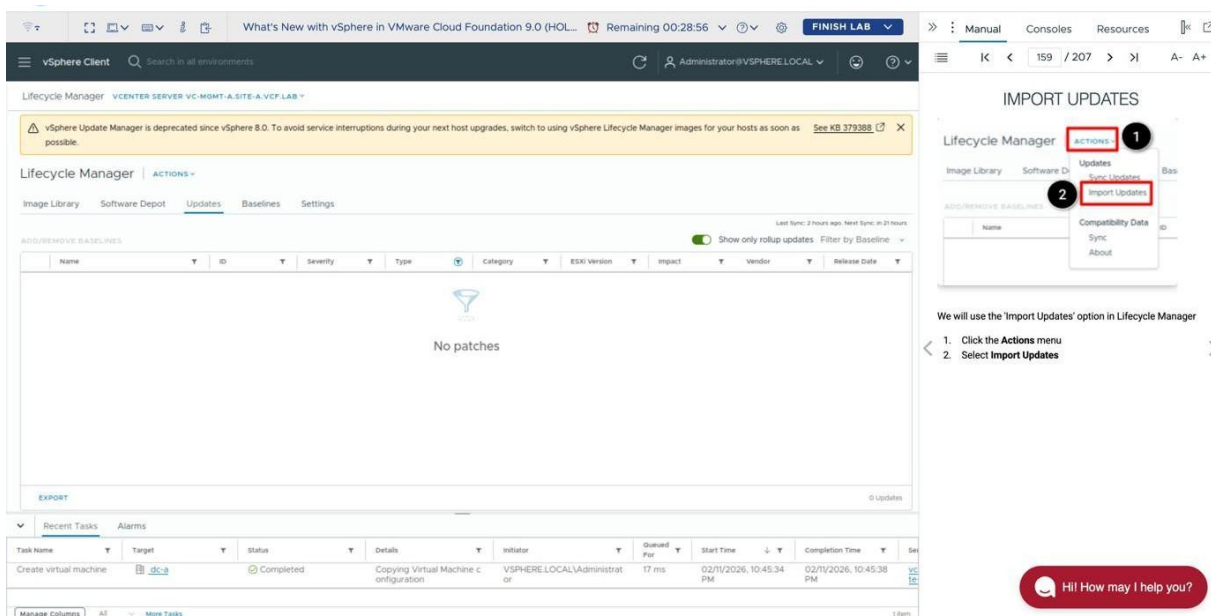
CLONE VTPM01

Now that we have a virtual machine with a vTPM attached, let's see how we can clone this VM and maintain a vTPM on the clone

1. Right-Click the VM vTPM01
2. Select Clone
3. Select Clone to Virtual Machine...

Hi! How may I help you?

2.17 Importing Updates from Life cycle Manager



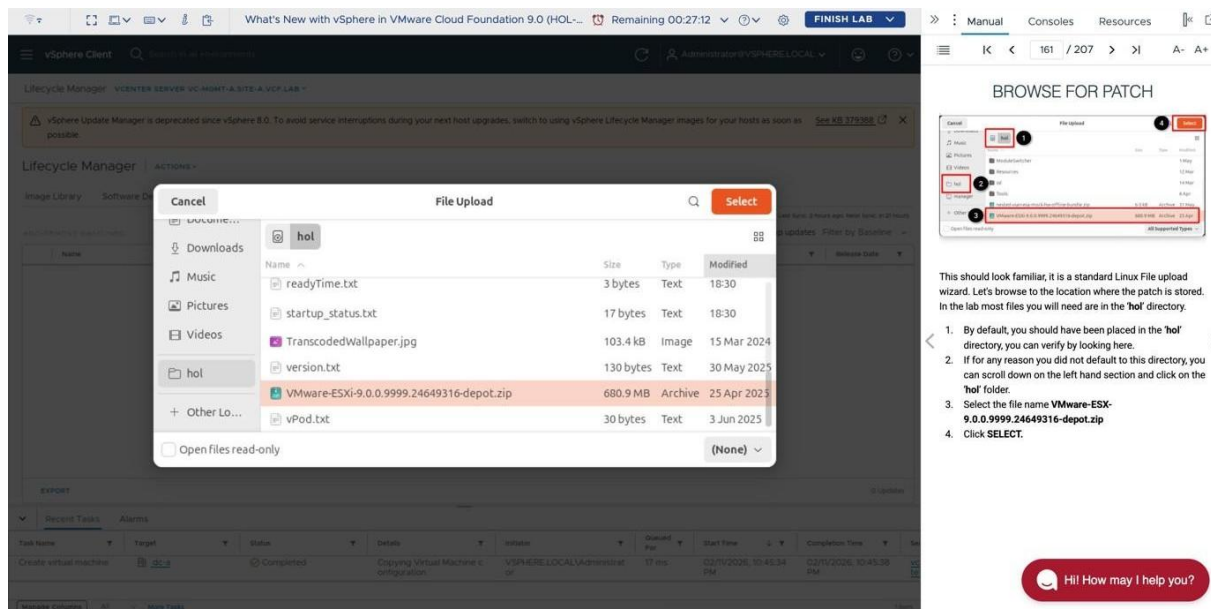
IMPORT UPDATES

We will use the 'Import Updates' option in Lifecycle Manager

1. Click the Actions menu
2. Select Import Updates

Hi! How may I help you?

2.18 Browsing for Patch on hol directory



File Upload

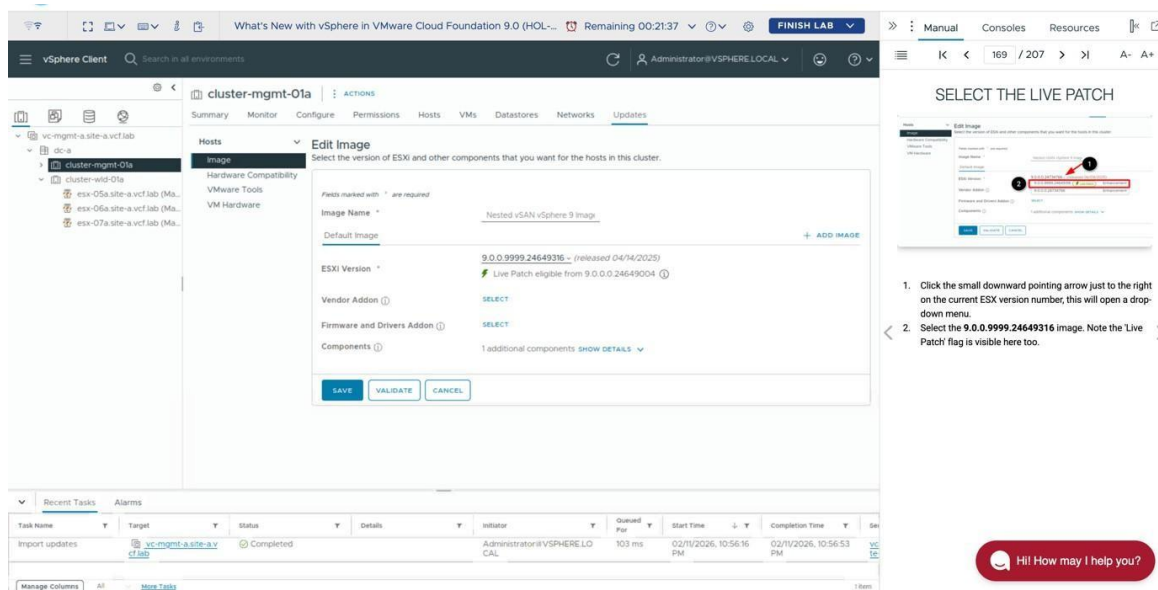
Name	Size	Type	Modified
readyTime.txt	3 bytes	Text	18:30
startup_status.txt	17 bytes	Text	18:30
TranscodedWallpaper.jpg	103.4 kB	Image	15 Mar 2024
version.txt	130 bytes	Text	30 May 2025
VMware-ESXi-9.0.0.9999.24649316-depot.zip	680.9 MB	Archive	25 Apr 2025
vPod.txt	30 bytes	Text	3 Jun 2025

BROWSE FOR PATCH

- By default, you should have been placed in the 'hol' directory, you can verify by looking here.
- If for any reason you did not default to this directory, you can scroll down on the left hand section and click on the 'hol' folder.
- Select the file name **VMware-ESXi-9.0.0.9999.24649316-depot.zip**
- Click **SELECT**.

Hi! How may I help you?

2.19 Selecting a live Patch



Edit Image

Select the version of ESXi and other components that you want for the hosts in this cluster.

Fields marked with * are required

Image Name: Nested vSAN vSphere 9 Image

ESXi Version: 9.0.0.9999.24649316 (released 04/24/2025)
Live Patch eligible from 9.0.0.0.24649004

Vendor Addon: SELECT

Firmware and Drivers Addon: SELECT

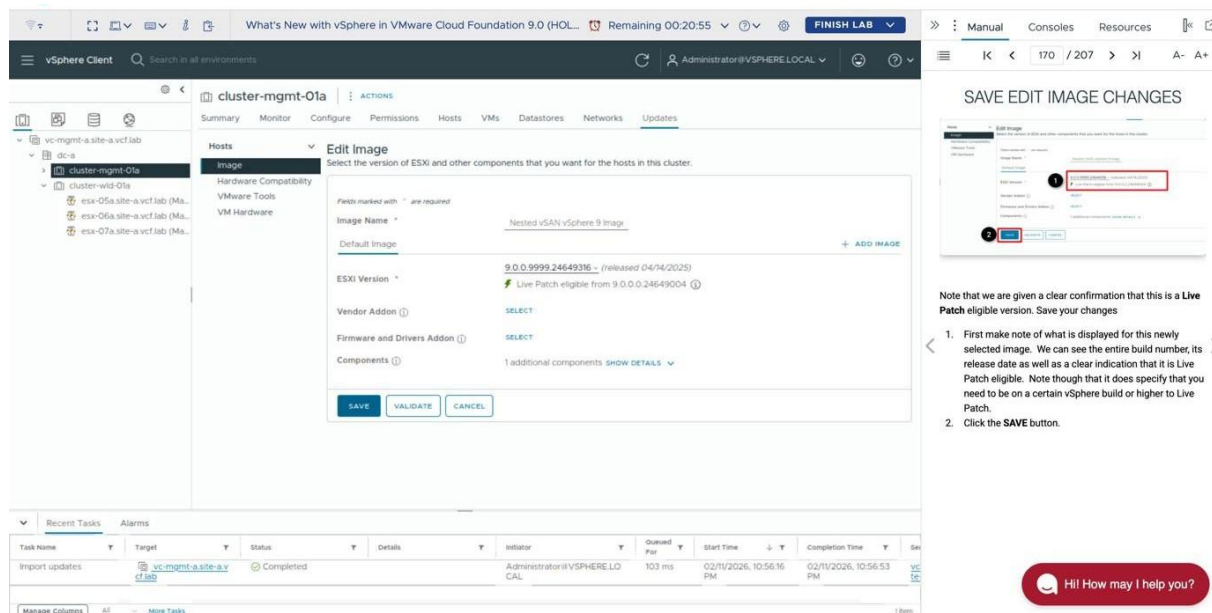
Components: 1 additional components SHOW DETAILS

SELECT THE LIVE PATCH

- Click the small downward pointing arrow just to the right on the current ESXi version number, this will open a drop-down menu.
- Select the **9.0.0.9999.24649316** image. Note the 'Live Patch' flag is visible here too.

Hi! How may I help you?

Saving Changes to Image



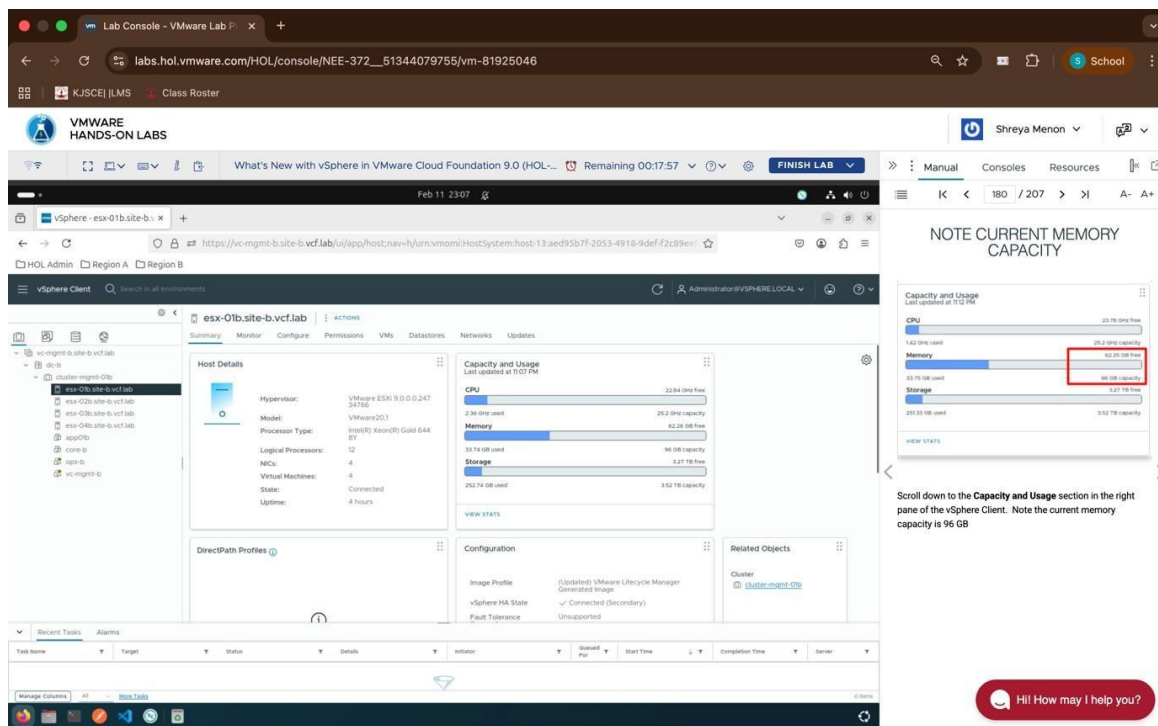
The screenshot shows the vSphere Client interface with the 'Edit Image' dialog open for the 'cluster-mgmt-01a' cluster. The dialog allows selecting an image, ESXi version, and vendor. A side note titled 'SAVE EDIT IMAGE CHANGES' provides instructions on how to save changes to a Live Patch eligible version.

SAVE EDIT IMAGE CHANGES

Note that we are given a clear confirmation that this is a **Live Patch** eligible version. Save your changes

1. First make note of what is displayed for this newly selected image. We can see the entire build number, its release date as well as a clear indication that it is Live Patch eligible. Note though that it does specify that you need to be on a certain vSphere build or higher to Live Patch.
2. Click the **SAVE** button.

2.20 Checking Capacity and Usage of vSphere Client – 96 GB



The screenshot shows the vSphere Client interface with the 'Host Details' section selected for the 'esx-01b.site-b.vcf.lab' host. The 'Capacity and Usage' section is expanded, showing the current memory usage and capacity.

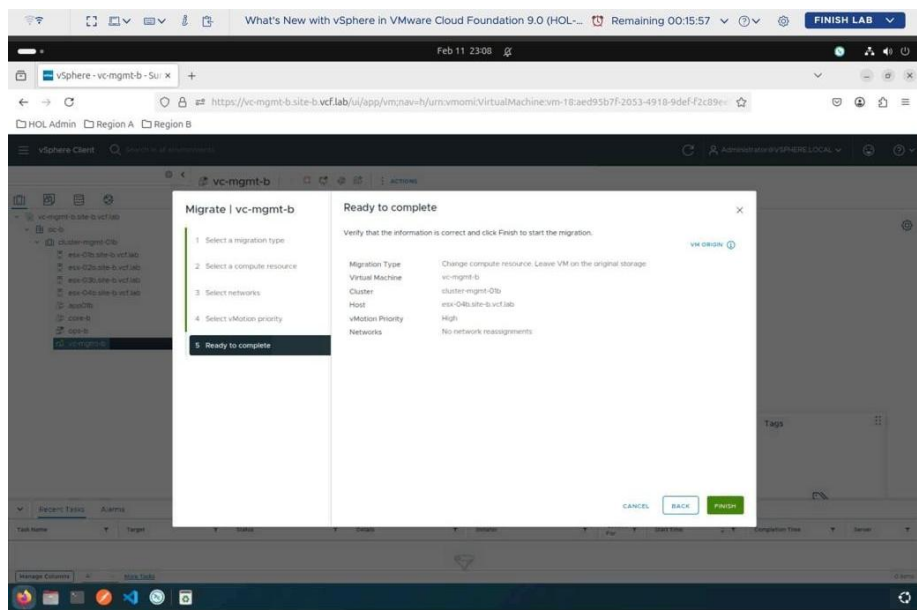
NOTE CURRENT MEMORY CAPACITY

Capacity and Usage
Last updated at 11:12 PM

Resource	Used	Free	Capacity
CPU	1.42 GHz used	25.78 GHz free	25.2 GHz capacity
Memory	33.75 GB used	62.25 GB free	96 GB capacity
Storage	251.33 GB used	2.27 TB free	3.02 TB capacity

Scroll down to the **Capacity and Usage** section in the right pane of the vSphere Client. Note the current memory capacity is 96 GB

2.21 Migrate Vmotion VC VM to another ESX Host

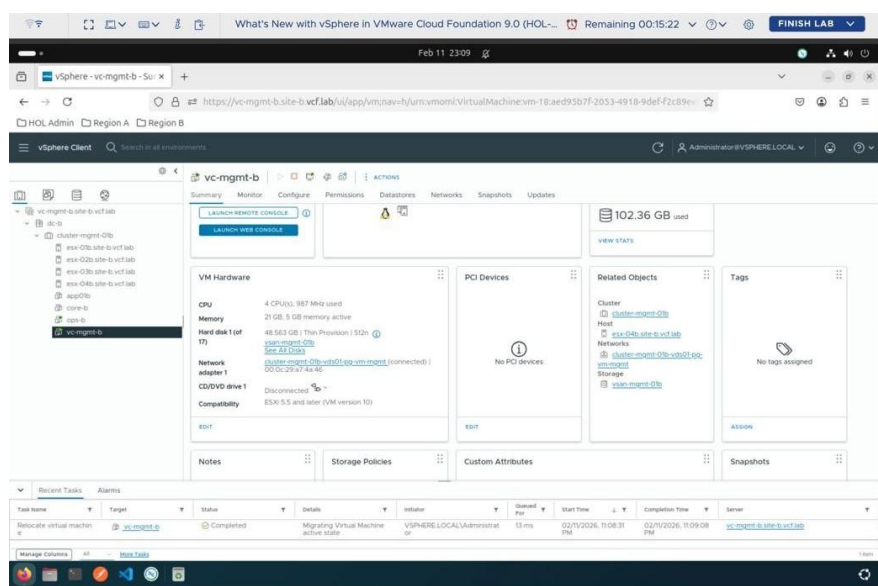


The screenshot shows the vSphere Client interface with the 'Ready to complete' migration wizard for VM 'vc-mgmt-b'. The wizard is titled 'Migrate | vc-mgmt-b' and 'Ready to complete'. It lists the migration details: Migration Type (Change compute resource), Virtual Machine (vc-mgmt-b), Cluster (cluster-mgmt-01b), Host (esx-04b.site-b.vcf.lab), vMotion Priority (High), and Networks (No network reassignments). The 'FINISH LAB' button is visible in the top right corner.

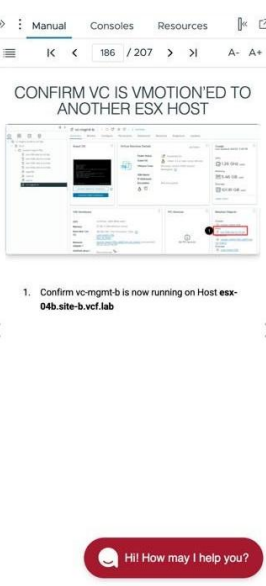


The screenshot shows the 'Ready to complete' migration wizard with the 'FINISH' button highlighted. The wizard is titled 'Migrate | vc-mgmt-b' and 'Ready to complete'. It lists the migration details: Migration Type (Change compute resource), Virtual Machine (vc-mgmt-b), Cluster (cluster-mgmt-01b), Host (esx-04b.site-b.vcf.lab), vMotion Priority (High), and Networks (No network reassignments). The 'FINISH LAB' button is visible in the top right corner.

Confirmation of VMOTION



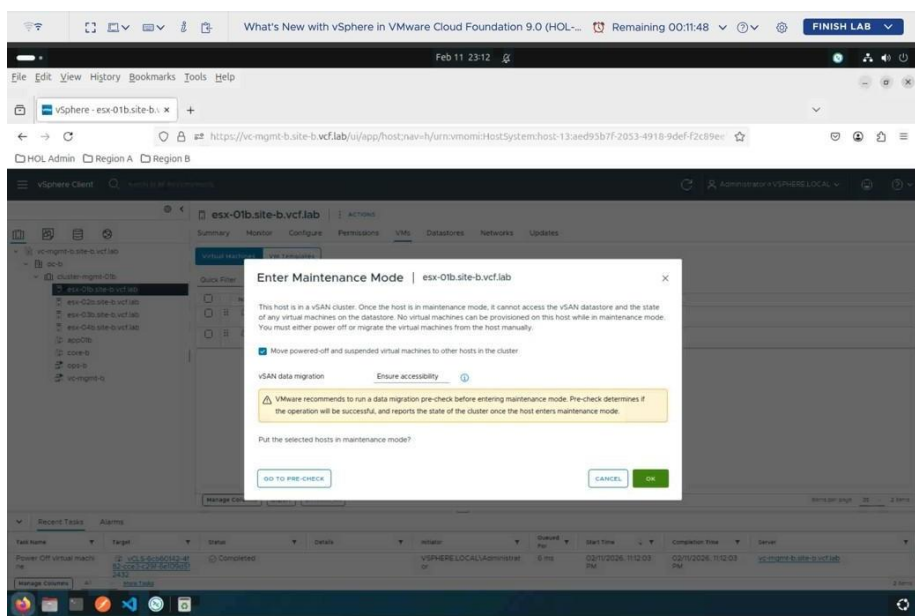
The screenshot shows the vSphere Client interface with the 'Summary' page for VM 'vc-mgmt-b'. The page displays various details about the VM, including its hardware, network, and storage. The 'Summary' tab is selected, showing the VM's name, ID, and various statistics. The 'FINISH LAB' button is visible in the top right corner.



The screenshot shows the 'Summary' page for VM 'vc-mgmt-b' with the 'FINISH' button highlighted. The page displays various details about the VM, including its hardware, network, and storage. The 'FINISH LAB' button is visible in the top right corner.

2.22 Enter Maintenance Mode

2.23 List Storage devices on Terminal



ENTER MAINTENANCE MODE (CONT.)

Enter Maintenance Mode | esx-01b.site-b.vcf.lab

This host is in a vSAN cluster. Once the host is in maintenance mode, it cannot access the vSAN datastore and the state of any virtual machines on the datastore. No virtual machines can be provisioned on this host while in maintenance mode. You must either power off or migrate the virtual machines from the host manually.

☒ Move powered-off and suspended virtual machines to other hosts in the cluster.

VSAN data migration: Ensure accessibility

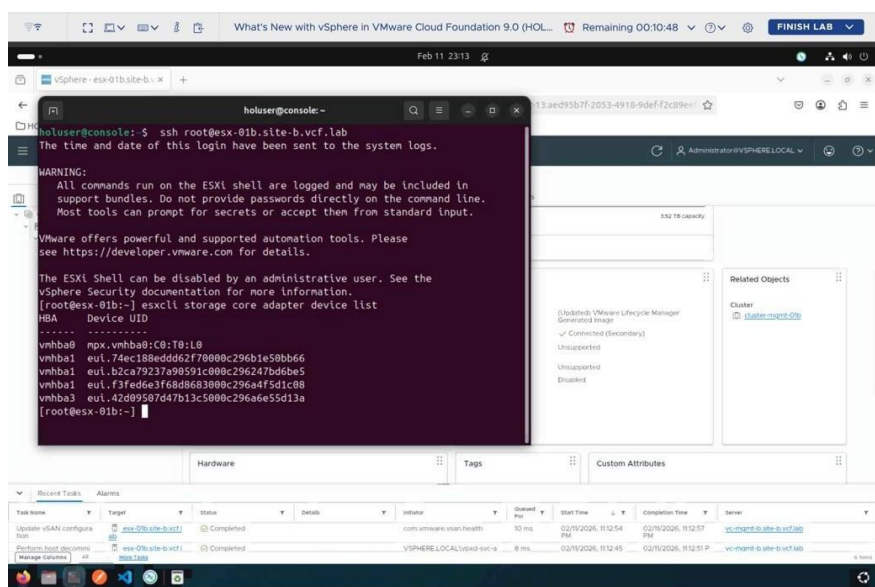
VMware recommends to run a data migration pre-check before entering maintenance mode. Pre-check determines if the operation will be successful, and reports the state of the cluster once the host enters maintenance mode.

Put the selected hosts in maintenance mode?

1. Confirm Move powered-off and suspended VMs... box is checked
2. Click OK

In case, new vCLS VM is created, Host Enter Maintenance mode will be in waiting state. Please follow the previous step to power off the newly created vCLS VM to complete the host maintenance mode step.

Hi! How may I help you?



LIST STORAGE DEVICES

holuser@console: ~

holuser@console: \$ ssh root@esx-01b.site-b.vcf.lab

The time and date of this login have been sent to the system logs.

WARNING:

All commands run on the ESXi shell are logged and may be included in support bundles. Do not provide passwords directly on the command line. Most tools can prompt for secrets or accept them from standard input.

VMware offers powerful and supported automation tools. Please see <https://developer.vmware.com> for details.

The ESXi Shell can be disabled by an administrative user. See the vSphere Security documentation for more information.

[root@esx-01b:~] esxcli storage core adapter device list

HBA Device UID

```

-----
vmhba0 npx.vmhba0:C0:T0:L0
vmhba1 eui.74ec188edd62f78000c296b1e50bb66
vmhba1 eui.b2ca79237f90591c000c296247b6db65
vmhba1 eui.f3fede63f6d8682000c2964ef5d1c08
vmhba3 eui.42d89507d47b13c5000c296a6e55d13a
[root@esx-01b:~]

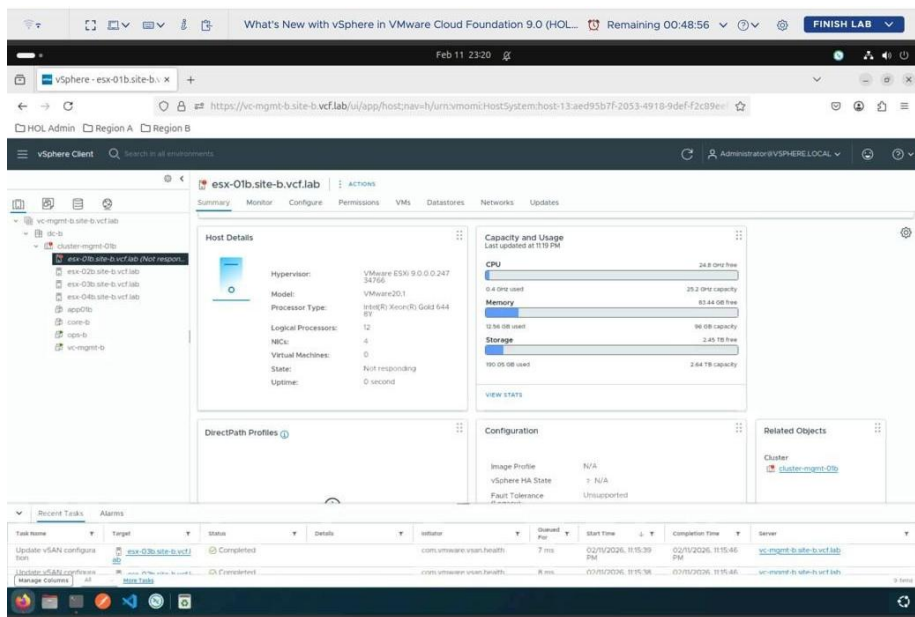
```

In order to enable the NVMe drive for memory tiering, we first need to figure out the Device UID of it. To do this, we will use an ESXCLI command to query all the adapter devices. In our lab, we have added the 6 GB NVMe to HBA3.

1. Type the following command in the terminal session
2. Take note of the device attached to vmhba3. In this case the NVMe device is `eui577142b0dffa0000c29646bba385f`. However, the UID for your lab may be different as this UID is generated as the lab is started. The important part is that you note the UID for vmhba3.

Hi! How may I help you?

2.24 Change Memory Capacity via Terminal to 102 GB

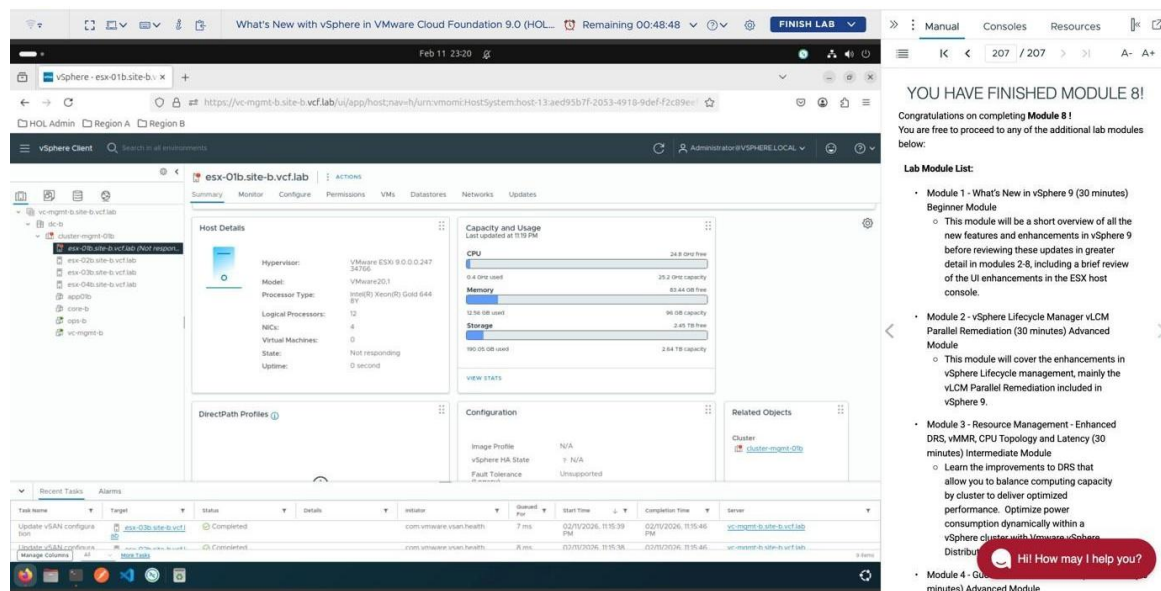


The screenshot shows the vSphere Client interface for the host **esx-01b.site-b.vcf.lab**. The **Capacity and Usage** section displays the following metrics:

Resource	Used	Capacity
CPU	0.4 GHz used	25.2 GHz free
Memory	12.37 GB used	102 GB free
Storage	100.08 GB used	2.45 TB free

The **Memory** row is highlighted with a red box, indicating the new capacity of 102 GB. Below the main interface, a task list shows the completion of the memory update.

2.25 Lab Completion



The screenshot shows the vSphere Client interface with a confirmation message: **YOU HAVE FINISHED MODULE 8!** Congratulations on completing Module 8! You are free to proceed to any of the additional lab modules below.

Lab Module List:

- Module 1 - What's New in vSphere 9 (30 minutes) Beginner Module**
 - This module will be a short overview of all the new features and enhancements in vSphere 9 before reviewing these updates in greater detail in modules 2-8, including a brief review of the UI enhancements in the ESX host console.
- Module 2 - vSphere Lifecycle Manager vLCM Parallel Remediation (30 minutes) Advanced Module**
 - This module will cover the enhancements in vSphere Lifecycle management, mainly the vLCM Parallel Remediation included in vSphere 9.
- Module 3 - Resource Management - Enhanced DRS, vMMR, CPU Topology and Latency (30 minutes) Intermediate Module**
 - Learn the improvements to DRS that allow you to balance computing capacity by cluster to deliver optimized performance. Optimize power consumption dynamically within a vSphere cluster with VMware vSphere Distributed Power Management.
- Module 4 - vSphere vMotion (30 minutes) Advanced Module**

Certificate

Certificate of Completion

Shreyans Tatiya

Has Successfully Completed the following Hands-on Lab:

**What's New with vSphere in VMware Cloud Foundation 9.0
(HOL-2630-01-VCF-L)**

Want to explore all the exciting new features, enhancements and updates available to you in the latest release of vSphere, vSphere 9.0? Then this lab is the right place to start! In this lab we will be reviewing areas such as the new licensing process which simplifies and streamlines license management with the new Unified license. We will also explore Memory Tiering and learn how the use of NVMe drives can enhance and optimize the amount of memory available to ESX hosts. In the Live Patch module you will learn how to patch your hosts with zero down time, improving and speeding up patching cycles. We will look at Multi-Vendor Clusters and how to manage the lifecycle of a cluster where you have more than one hardware vendor's systems together. We will also look at the enhancements and improvements to Lifecycle Manager and how you can minimize your administration requirements through the use of cluster level images. And much more..

Start Date Time: 05:50 AM 02/12/2026
Completion Date Time: 07:21 AM 02/12/2026
Time Spent: 89 minutes
Language: en

Conclusion:

This lab provided a comprehensive overview of the latest enhancements in VMware vSphere highlighting improvements in licensing, performance, lifecycle management, and operational efficiency.

Key features such as the Unified License model, Memory Tiering with NVMe, Live Patch

Department of Computer Engineering

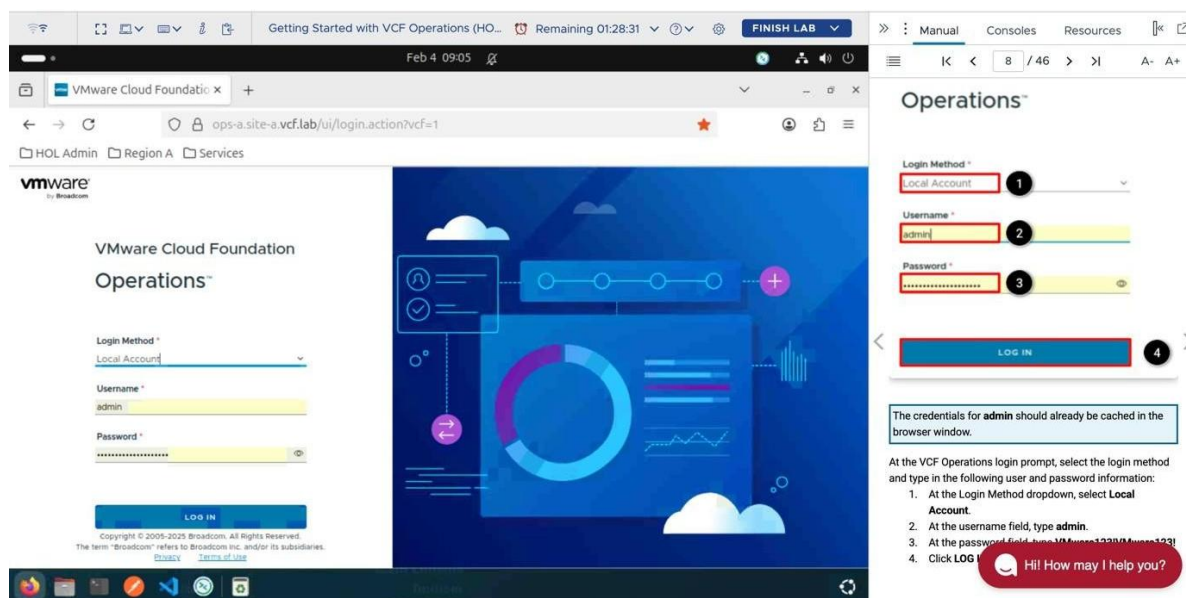
for zero-downtime updates, Multi-Vendor Clusters, and Lifecycle Manager enhancements demonstrate how vSphere 9.0 simplifies administration while increasing availability and scalability.

Part B

Getting Started with VCF Operations (HOL-2601-01-VCF-L)

Steps Performed

1.1 Login to VCF Operations Console



Getting Started with VCF Operations (HO... Remaining 01:28:31 FINISH LAB

Feb 4 09:05

ops-a.site-a.vcf.lab/ui/login.action?vcf=1

HOL Admin Region A Services

vmware
by Broadcom

VMware Cloud Foundation
Operations™

Login Method *
Local Account

Username *
admin

Password *

LOG IN

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The term "Broadcom" refers to Broadcom Inc. and/or its subsidiaries.
[Privacy](#) [Terms of Use](#)

Manual Consoles Resources

Operations™

Login Method *
Local Account

Username *
admin

Password *

LOG IN

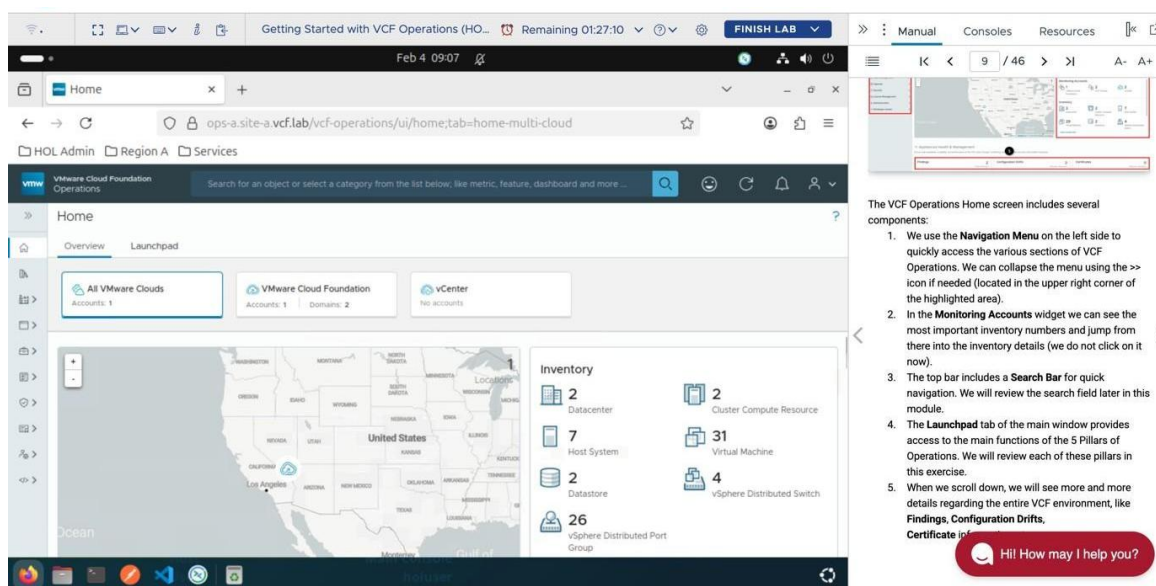
The credentials for **admin** should already be cached in the browser window.

At the VCF Operations login prompt, select the login method and type in the following user and password information:

1. At the Login Method dropdown, select **Local Account**.
2. At the username field, type **admin**.
3. At the password field, type **admin12345678910**.
4. Click **LOG IN**.

Hi! How may I help you?

1.2 VCF Operations Home Screen



Getting Started with VCF Operations (HO... Remaining 01:27:10 FINISH LAB

Feb 4 09:07

ops-a.site-a.vcf.lab/vcf-operations/ui/home;tab=home-multi-cloud

HOL Admin Region A Services

vmware
by Broadcom

VMware Cloud Foundation
Operations™

Search for an object or select a category from the list below, like metric, feature, dashboard and more...

Home

Overview Launchpad

All VMware Clouds
Accounts: 1

VMware Cloud Foundation
Accounts: 1 Domains: 2

vCenter
No accounts

Inventory

2 Datacenter

7 Host System

2 Datastore

26 vSphere Distributed Port Group

2 Cluster Compute Resource

31 Virtual Machine

4 vSphere Distributed Switch

Manual Consoles Resources

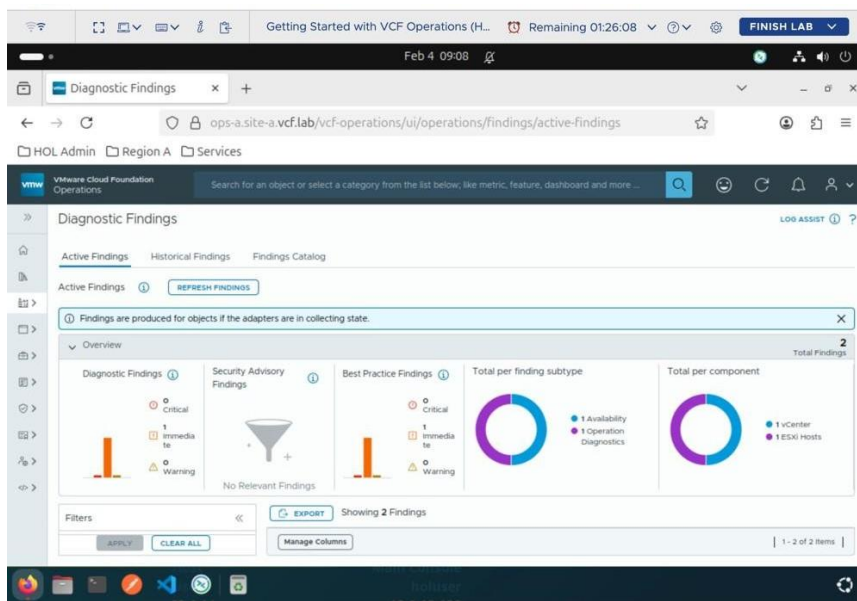
The VCF Operations Home screen includes several components:

1. We use the **Navigation Menu** on the left side to quickly access the various sections of VCF Operations. We can collapse the menu using the >> icon if needed (located in the upper right corner of the highlighted area).
2. In the **Monitoring Accounts** widget we can see the most important inventory numbers and jump from there into the inventory details (we do not click on it now).
3. The top bar includes a **Search Bar** for quick navigation. We will review the search field later in this module.
4. The **Launchpad** tab of the main window provides access to the main functions of the 5 Pillars of Operations. We will review each of these pillars in this exercise.
5. When we scroll down, we will see more and more details regarding the entire VCF environment, like **Findings, Configuration Drifts, Certificate** in...

Hi! How may I help you?

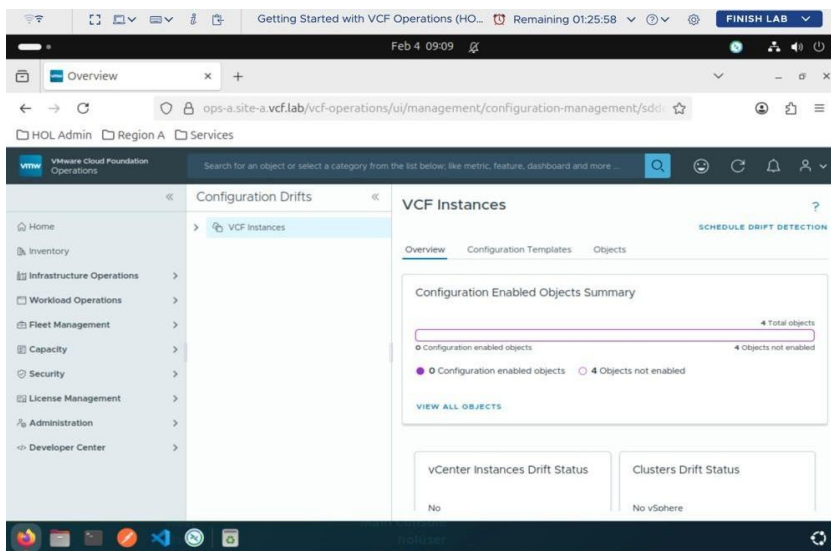
1.3 Findings Overview

In the left bottom, we see an overview of all the findings. You can see how many findings have the severity of critical, immediate or warning.



The screenshot shows the VMware Cloud Foundation Operations console. The left sidebar contains navigation links like Home, Inventory, Infrastructure Operations, Workload Operations, Fleet Management, Capacity, Security, License Management, Administration, and Developer Center. The main content area is titled 'Diagnostic Findings' and includes tabs for Active Findings, Historical Findings, and Findings Catalog. The 'Active Findings' tab is selected, showing a summary of findings with a bar chart and a funnel icon. The right sidebar displays a 'FINDINGS OVERVIEW' widget with a text description and a list of findings. Below the widget, there are instructions: '1. Scroll down until we see the Findings widget.' and '2. Click on the VIEW FINDINGS to get to the detailed Diagnostics findings dashboard.'

1.4 Configuration Drifts Overview

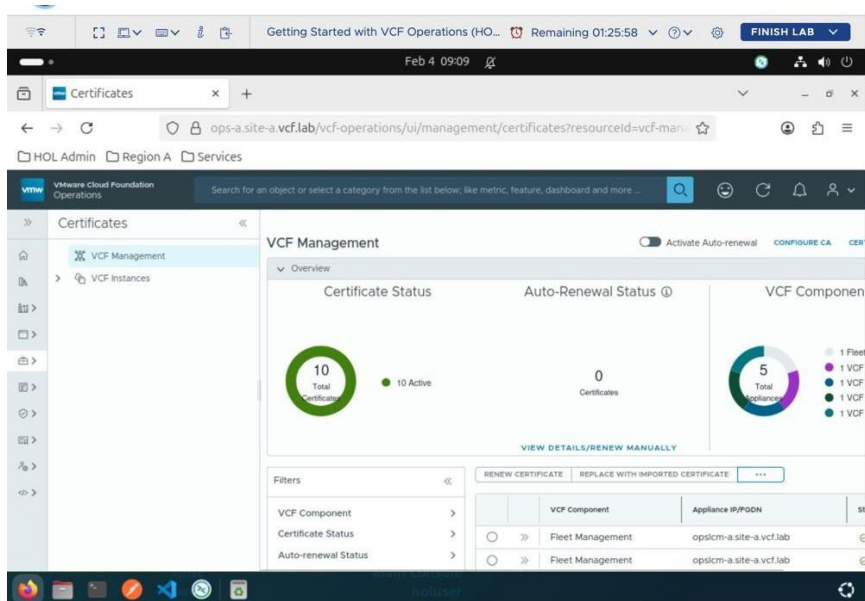


The screenshot shows the VMware Cloud Foundation Operations console. The left sidebar contains navigation links like Home, Inventory, Infrastructure Operations, Workload Operations, Fleet Management, Capacity, Security, License Management, Administration, and Developer Center. The main content area is titled 'Configuration Drifts' and includes tabs for Overview, Configuration Templates, and Objects. The 'Overview' tab is selected, showing a summary of configuration drifts with a bar chart and a funnel icon. The right sidebar displays a 'CONFIGURATION DRIFTS OVERVIEW' widget with a text description and a list of drifts. Below the widget, there are instructions: '1. Click on the View Drifts to get to the detailed dashboard.'

1.5 Certificates Overview

We can see if the certificates of all the appliances are correctly working or how many need to be replaced.

1.6 Workload Operations



The screenshot shows the VMware Cloud Foundation interface for 'Certificates Overview'. The main panel displays 'VCF Management' with a 'Certificate Status' gauge showing 10 Total Certificates (10 Active) and an 'Auto-Renewal Status' gauge showing 0 Certificates. A table below lists VCF Components and their Appliance IP/FQDNs.

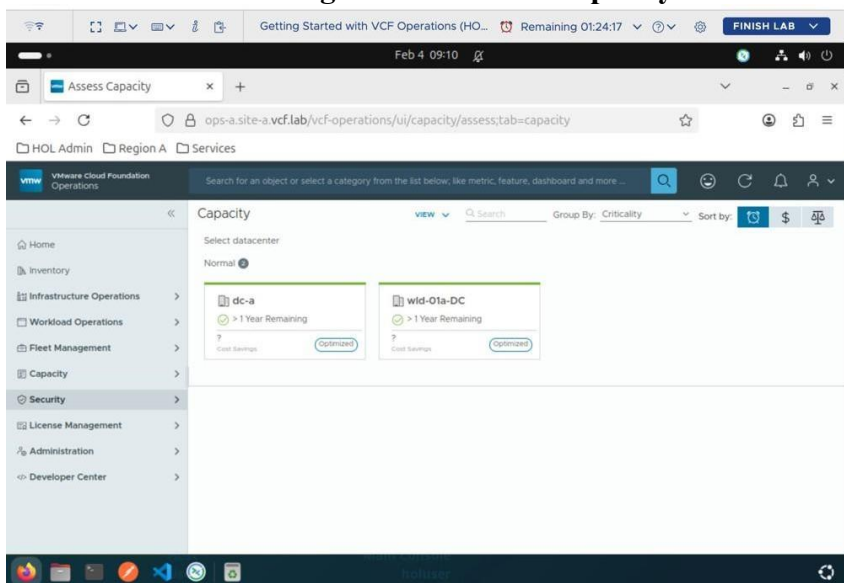
CERTIFICATES OVERVIEW

In the right pane, we have an overview of the status of all the certificates. We can see if the certificates of all the appliances are correctly working or how many need to be replaced.

1. Click on the **VIEW CERTIFICATES** to get to the detailed dashboard.

Hi! How may I help you?

VIEW CAPACITY to go to the detailed capacity dashboard



The screenshot shows the VMware Cloud Foundation interface for 'Capacity'. The main panel displays 'Capacity' with a 'Select datacenter' dropdown and two datacenter cards: 'dc-a' and 'wid-01a-DC', both showing '> 1 Year Remaining' and 'Optimized' status.

WORKLOAD OPERATIONS

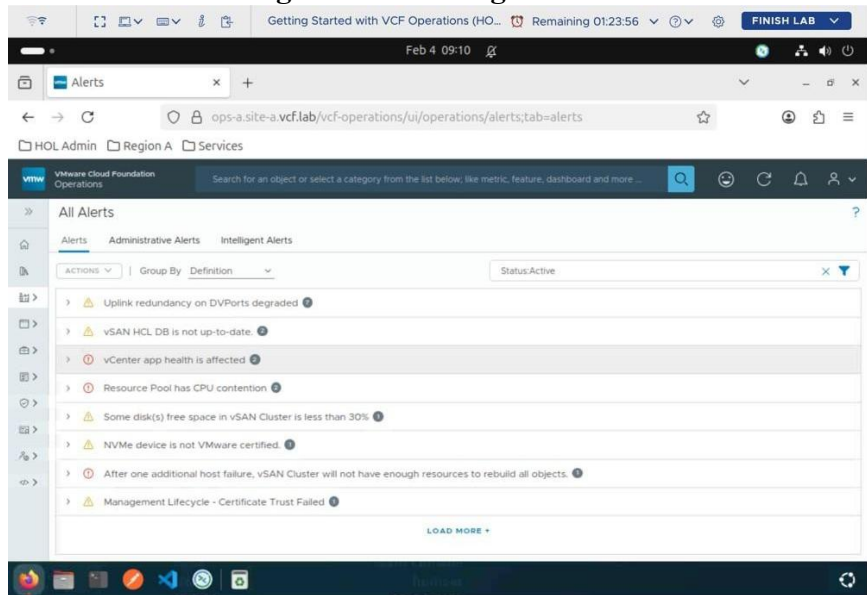
If we scroll down, we can find the **Workload Operations** widget. Here we find an overall view of your capacity and alerts. From here we can click on the **VIEW** links in every section to get to the details, if you do this, use the "Navigate Back to Home Screen" step or the back button in the browser to get back to Workload Operations.

1. Click on the **VIEW CAPACITY** to go to the detailed capacity dashboard.
2. Click on **VIEW ALERTS** to go to the alerting dashboard.
3. Click on **VIEW COSTS** to go to the cost dashboard.

Hi! How may I help you?

Department of Computer Engineering

VIEW ALERTS to go to the alerting dashboard



The screenshot shows the VMware Cloud Foundation Operations Alerts dashboard. The left sidebar contains navigation links for Alerts, Administrative Alerts, and Intelligent Alerts. The main area displays a list of alerts with details such as 'Uplink redundancy on DVP ports degraded', 'vSAN HCL DB is not up-to-date', and 'vCenter app health is affected'. A 'VIEW ALERTS' link is visible at the bottom of the list.

WORKLOAD OPERATIONS

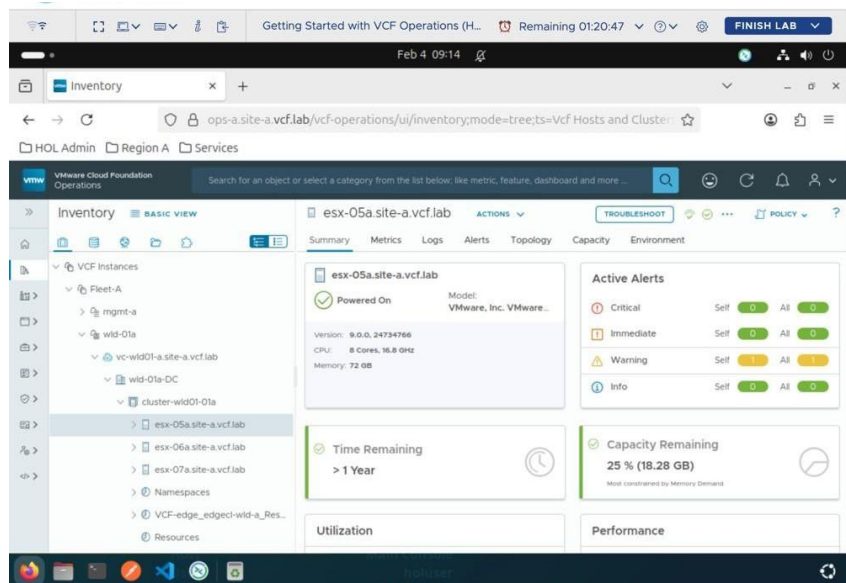
If we scroll down, we can find the **Workload Operations** widget. Here we find an overall view of your capacity and alerts. From here we can click on the **VIEW** links in every section to get to the details, if you do this, use the "Navigate Back to Home Screen" step or the back button in the browser to get back to Workload Operations.

1. Click on the **VIEW CAPACITY** to go to the detailed capacity dashboard.
2. Click on **VIEW ALERTS** to go to the alerting dashboard.
3. Click on **VIEW COSTS** to go to the cost dashboard.

Hi! How may I help you?

1.7 Using the Inventory

The Inventory is a collection of all objects discovered and monitored by VCF Operations.



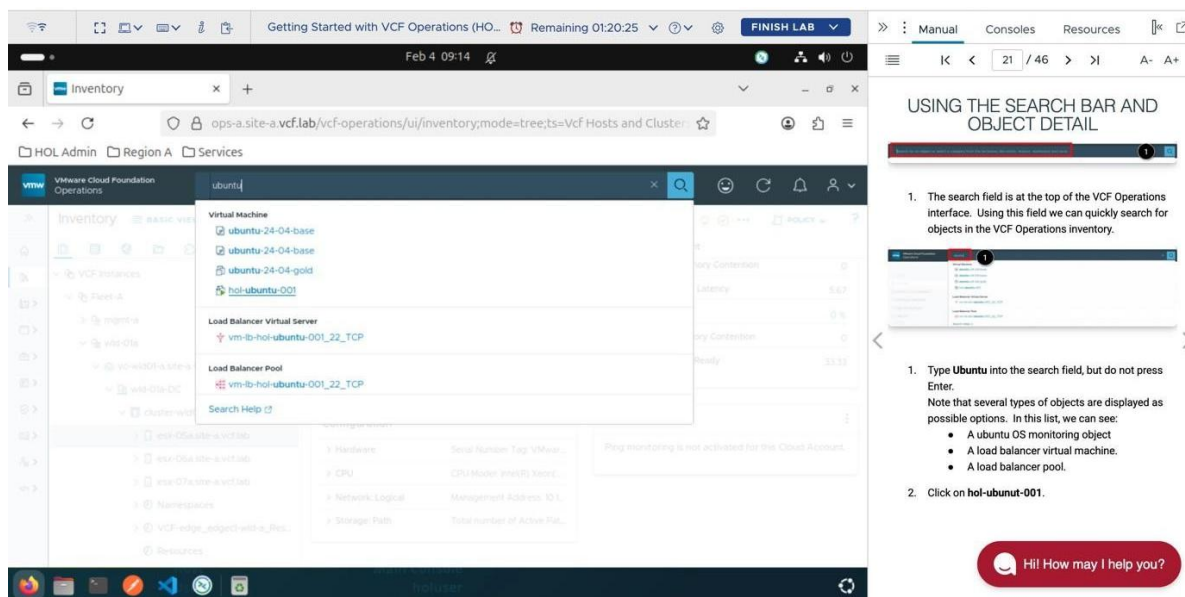
The screenshot shows the VMware Cloud Foundation Operations Inventory dashboard. The left sidebar contains navigation links for Inventory, BASIC VIEW, and esx-05a.site-a.vcf.lab. The main area displays a tree view of the inventory with details such as 'Powered On', 'Model: VMware, Inc. VMware...', 'Version: 9.0.0.24734766', 'CPU: 8 Cores, 16.8 GHz', and 'Memory: 72 GB'. A 'VIEW ALERTS' link is visible at the bottom of the list.

USING THE INVENTORY

The Inventory is a collection of all objects discovered and monitored by VCF Operations. We can navigate this inventory in a tree view similar to the one found in the vSphere Client.

1. Under VCF Instances, click the > next to Fleet-A to expand the inventory.
2. Click the > next to wld-01a to expand the inventory for this workload cluster.
3. Click on the > next to vc-wld01-a.site-a.vcf.lab to expand the inventory of the vCenter.
4. Click on the > next to wld-01a-DC to expand the inventory of the virtual Datacenter.
5. Click on the > next to cluster-wld01-01a to expand the inventory of the Cluster.
6. Click on one of the ESXi hosts, esxi-05a.site-a.vcf.lab in this example. This will open the object view for this.

Hi! How may I help you?



Getting Started with VCF Operations (HOL) Remaining 01:20:25 FINISH LAB

Feb 4 09:14

Inventory

ops.site-a.vcf.lab/vcf-operations/ui/inventory;mode=tree;ts=Vcf Hosts and Cluster

HOL Admin Region A Services

Virtual Machine

- ubuntu-24-04-base
- ubuntu-24-04-gold
- hol-ubuntu-001

Load Balancer Virtual Server

- vm-lb-hol-ubuntu-001_22_TCP

Load Balancer Pool

- vm-lb-hol-ubuntu-001_22_TCP

Search Help

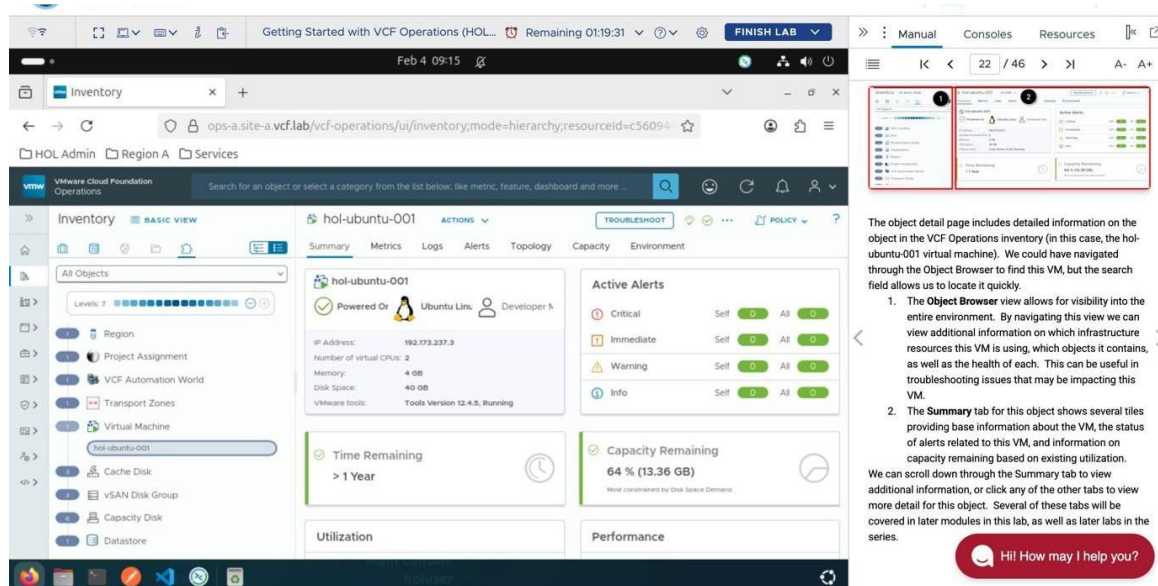
Using the Search Bar and Object Detail

- The search field is at the top of the VCF Operations interface. Using this field we can quickly search for objects in the VCF Operations inventory.
- Type **Ubuntu** into the search field, but do not press Enter. Note that several types of objects are displayed as possible options. In this list, we can see:
 - A ubuntu OS monitoring object
 - A load balancer virtual machine.
 - A load balancer pool.

Hi! How may I help you?

1.8 Object Detail Page

The object detail page includes detailed information on the object in the VCF Operations inventory (in this case, the hol-ubuntu-001 virtual machine).



Getting Started with VCF Operations (HOL) Remaining 01:19:31 FINISH LAB

Feb 4 09:15

Inventory

ops.site-a.vcf.lab/vcf-operations/ui/inventory;mode=hierarchy;resourceId=c56094

HOL Admin Region A Services

Search for an object or select a category from the list below, like metric, feature, dashboard and more...

Inventory BASIC VIEW

hol-ubuntu-001

Summary Metrics Logs Alerts Topology Capacity Environment

Summary

hol-ubuntu-001

Powered On Ubuntu Linux Developer k

IP Address: 192.173.237.3

Number of virtual CPUs: 2

Memory: 4 GB

Disk Space: 40 GB

VMware tools: Tools Version 12.4.5, Running

Active Alerts

Alert	Severity	Self	All
Critical	Self	0	0
Immediate	Self	0	0
Warning	Self	0	0
Info	Self	0	0

Time Remaining > 1 Year

Capacity Remaining 64 % (13.36 GB)

Utilization

Performance

The object detail page includes detailed information on the object in the VCF Operations inventory (in this case, the hol-ubuntu-001 virtual machine). We could have navigated through the Object Browser to find this VM, but the search field allows us to locate it quickly.

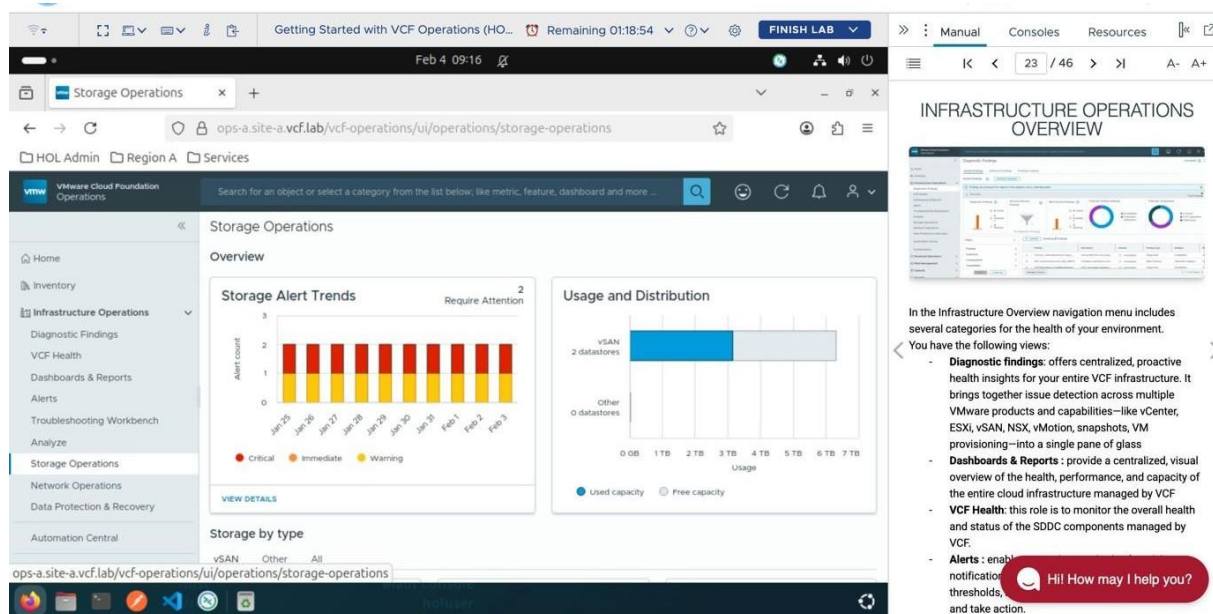
- The **Object Browser** view allows for visibility into the entire environment. By navigating this view we can view additional information on which infrastructure resources this VM is using, which objects it contains, as well as the health of each. This can be useful in troubleshooting issues that may be impacting this VM.
- The **Summary** tab for this object shows several tiles providing base information about the VM, the status of alerts related to this VM, and information on capacity remaining based on existing utilization. We can scroll down through the Summary tab to view additional information, or click any of the other tabs to view more detail for this object. Several of these tabs will be covered in later modules in this lab, as well as later labs in the series.

Hi! How may I help you?

Department of Computer Engineering

1.9 Infrastructure Operations Overview

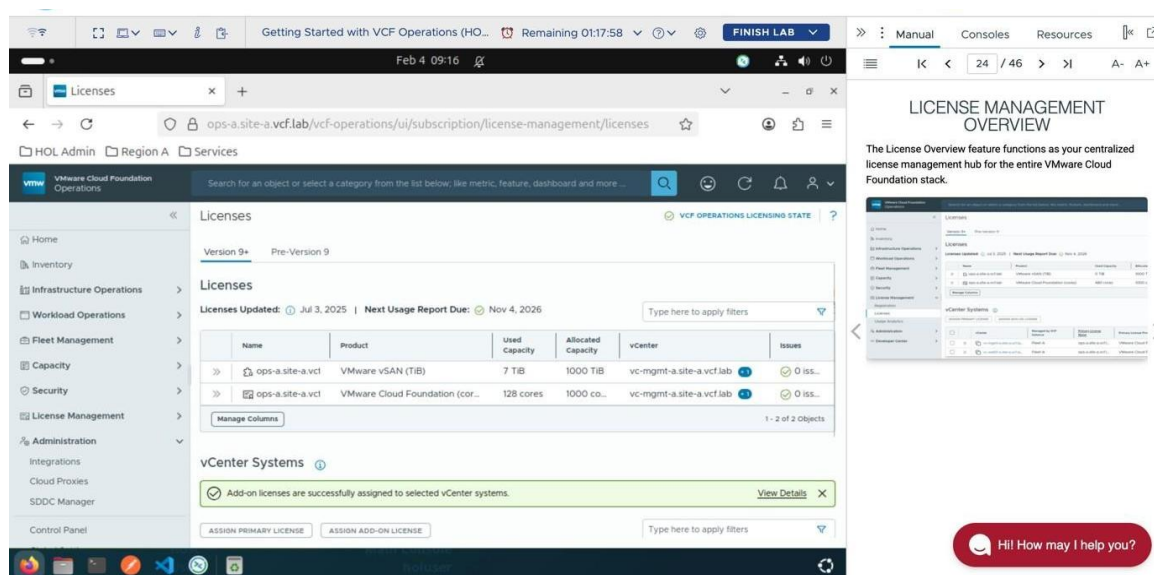
In the Infrastructure Overview navigation menu includes several categories for the health of your environment.



The screenshot shows the VMware Cloud Foundation Infrastructure Operations Overview dashboard. The left sidebar contains navigation options: Home, Inventory, Infrastructure Operations (selected), Diagnostic Findings, VCF Health, Dashboards & Reports, Alerts, Troubleshooting Workbench, Analyze, Storage Operations, Network Operations, Data Protection & Recovery, and Automation Central. The main content area displays 'Storage Operations' with an 'Overview' section. It includes a 'Storage Alert Trends' bar chart showing alert counts over time, categorized by Critical (red), Immediate (orange), and Warning (yellow). A 'Usage and Distribution' bar chart shows usage for vSAN and Other datastores. A 'Storage by type' section shows vSAN and Other storage types. The right sidebar shows a 'Manual' tab and a 'Resources' tab. Below the main content area, there is a 'Hi! How may I help you?' chat bubble.

1.10 License Management Overview

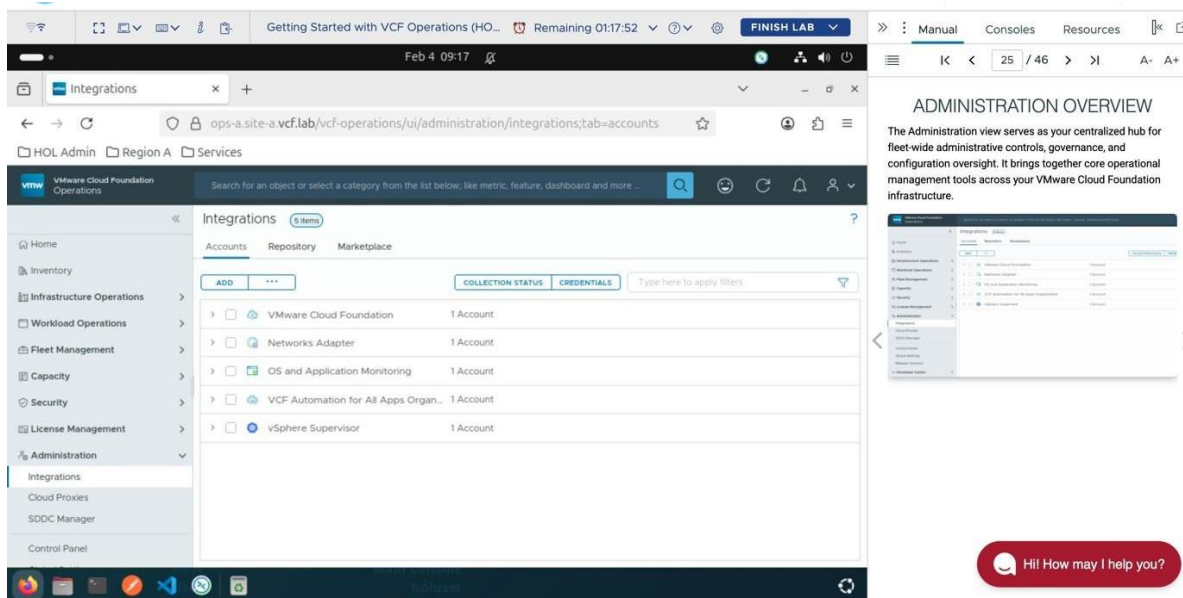
The License Overview feature functions as your centralized license management hub for the entire VMware Cloud Foundation stack.



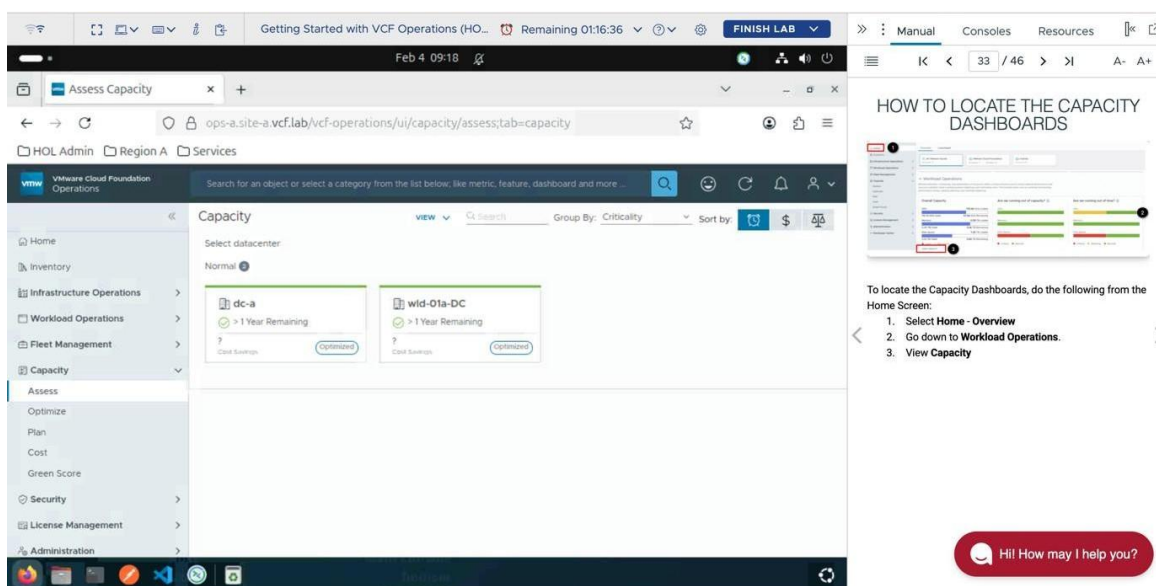
The screenshot shows the VMware Cloud Foundation License Management Overview dashboard. The left sidebar contains navigation options: Home, Inventory, Infrastructure Operations, Workload Operations, Fleet Management, Capacity, Security, License Management (selected), Administration, Integrations, Cloud Proxies, SDDC Manager, and Control Panel. The main content area displays 'Licenses' with a 'Version 9+' and 'Pre-Version 9' section. It includes a 'Licenses Updated' section showing the next usage report due date. A table lists licenses with columns: Name, Product, Used Capacity, Allocated Capacity, vCenter, and Issues. The table shows two licenses: 'ops-a-site-a-vc1' (VMware vSAN (TiB)) and 'ops-a-site-a-vc1' (VMware Cloud Foundation (cor...)). Below the table, there is a 'vCenter Systems' section showing a message: 'Add-on licenses are successfully assigned to selected vCenter systems.' The right sidebar shows a 'Manual' tab and a 'Resources' tab. Below the main content area, there is a 'Hi! How may I help you?' chat bubble.

1.11 Administration overview

The Administration view serves as your centralized hub for fleet-wide administrative controls, governance, and configuration oversight. It brings together core operational management tools across your VMware Cloud Foundation infrastructure.

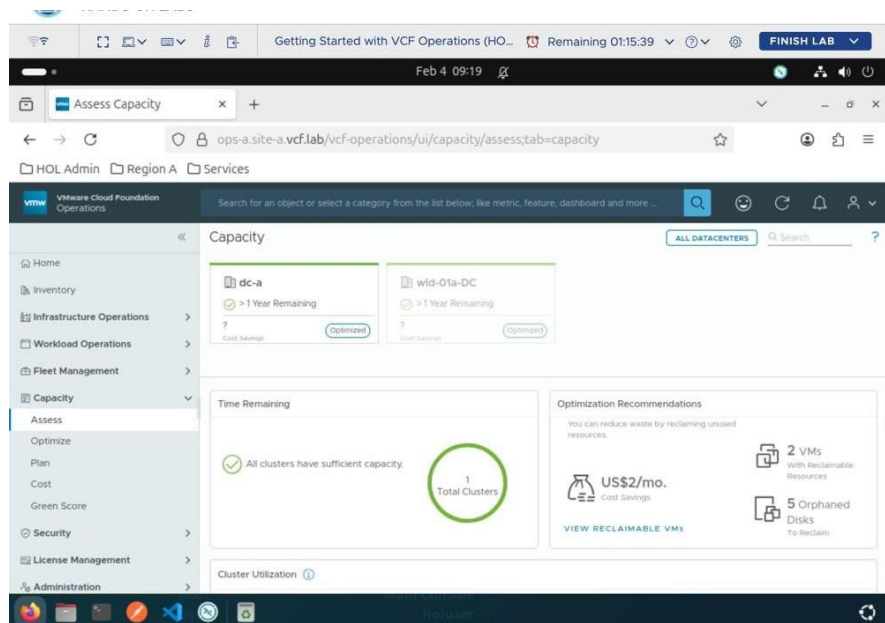


1.12 Capacity Dashboards



1.13 Capacity Overview

In the Capacity navigation menu,, there are five main components: Assess, Optimize, and Plan, Cost and Green score.

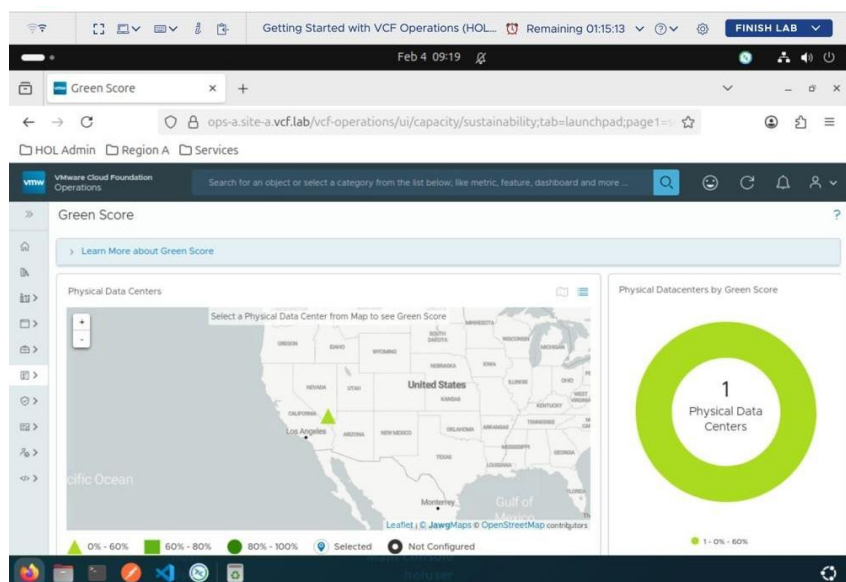


The screenshot shows the VMware Cloud Foundation Capacity Overview dashboard. The left sidebar contains navigation links: Home, Inventory, Infrastructure Operations, Workload Operations, Fleet Management, Capacity, Assess, Optimize, Plan, Cost, Green Score, Security, License Management, and Administration. The main content area displays 'Capacity' information for two datacenters: 'dc-a' and 'wid-01a-DC'. It shows 'Time Remaining' for each, with 'dc-a' having '> 1 Year Remaining' and 'wid-01a-DC' having '> 1 Year Remaining'. A 'Cluster Utilization' section indicates 'All clusters have sufficient capacity' and '1 Total Clusters'. An 'Optimization Recommendations' section shows 'US\$2/mo. Cost Savings' and '2 VMs With Reclaimable Resources'. A 'VIEW RECLAIMABLE VMs' button is present. A 'Cluster Utilization' link is at the bottom.

In the Capacity navigation menu, there are five main components: Assess, Optimize, and Plan, Cost and Green score. We won't review each component in-depth in this lab, but below is a general overview of each:

1. Assess – Use this component to determine if there is sufficient capacity in our VMware Cloud environment for current and future workloads.
2. Optimize – Use this component to optimize our data center for optimal workload performance by ensuring our workloads have the resources they need (includes the Rightsize and Reclaim features).
3. Plan – Use this component to plan for the addition/removal of capacity or workloads in the environment, as well as evaluate the possibility of moving workloads across supported clouds.
4. Cost - Use this component to give an overview of your costs in the datacenter.
5. Green Score - Here you can see an overview of power consumption.

Hi! How may I help you?



The screenshot shows the VMware Cloud Foundation Green Score dashboard. The left sidebar contains navigation links: Home, Inventory, Infrastructure Operations, Workload Operations, Fleet Management, Capacity, Assess, Optimize, Plan, Cost, Green Score, Security, License Management, and Administration. The main content area displays 'Green Score' information. It includes a 'Learn More about Green Score' link. A 'Physical Data Centers' map shows the United States with a green triangle indicating a selected location. A 'Physical Datacenters by Green Score' section shows a green circle with '1 Physical Data Centers'. A legend at the bottom indicates '0% - 60%', '60% - 80%', '80% - 100%', 'Selected', and 'Not Configured'.

In the Capacity navigation menu, there are five main components: Assess, Optimize, and Plan, Cost and Green score. We won't review each component in-depth in this lab, but below is a general overview of each:

1. Assess – Use this component to determine if there is sufficient capacity in our VMware Cloud environment for current and future workloads.
2. Optimize – Use this component to optimize our data center for optimal workload performance by ensuring our workloads have the resources they need (includes the Rightsize and Reclaim features).
3. Plan – Use this component to plan for the addition/removal of capacity or workloads in the environment, as well as evaluate the possibility of moving workloads across supported clouds.
4. Cost - Use this component to give an overview of your costs in the datacenter.
5. Green Score - Here you can see an overview of power consumption and carbon footprint.

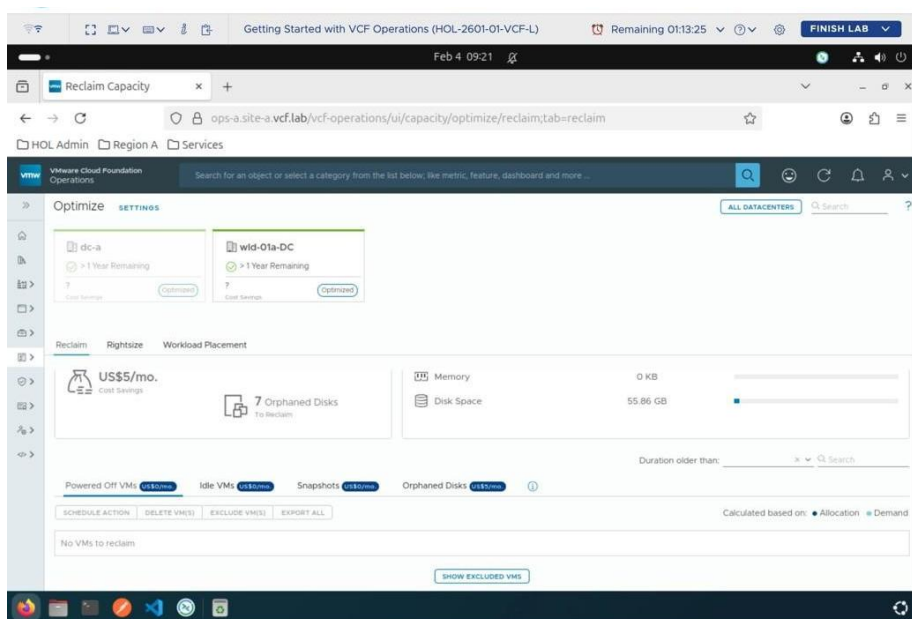
Hi! How may I help you?

Department of Computer Engineering

1.14 Datacenter Capacity Time Remaining and Optimization Recommendations

Time Remaining -- using advanced calculations, Aria Operations determines if any additional resources need to be added to maintain the current state of the environment.

Optimization Recommendations -- this component identifies available resources that could be reclaimed and provides an associated cost savings estimate.



Reclaim Capacity

dc-a: > 1 Year Remaining
wid-01a-DC: > 1 Year Remaining

Cost Savings: US\$5/mo.
7 Orphaned Disks

Reclaim Rightsize Workload Placement

Powered Off VMs Idle VMs Snapshots Orphaned Disks

NO VMs to reclaim

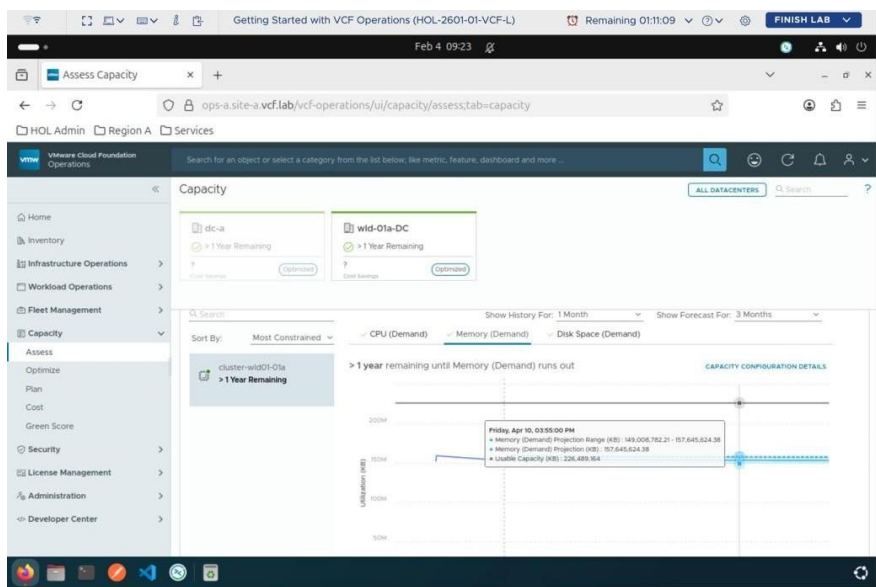
SHOW EXCLUDED VMs

DATACENTER CAPACITY TIME REMAINING AND OPTIMIZATION RECOMMENDATIONS

The first two components in the Datacenter Capacity window are Time Remaining and Optimization Recommendations. Below are brief overviews of both.

1. Time Remaining -- using advanced calculations, Aria Operations determines if any additional resources need to be added to maintain the current state of the environment. Note: Depending upon the available resources of the lab, we may see different results than what is shown in this example.
2. Optimization Recommendations -- this component identifies available resources that could be reclaimed and provides an associated cost savings estimate. Note: Depending upon the available resources of the lab, we may see different results than what is shown in this example.
3. Use the slide bar on the right side to scroll down and see other options which are discussed on the next page.

Hi! How may I help you?



Assess Capacity

dc-a: > 1 Year Remaining
wid-01a-DC: > 1 Year Remaining

Show History For: 1 Month Show Forecast For: 3 Months

Sort By: Most Constrained

cluster-wid01-01a: > 1 Year Remaining

> 1 year remaining until Memory (Demand) runs out

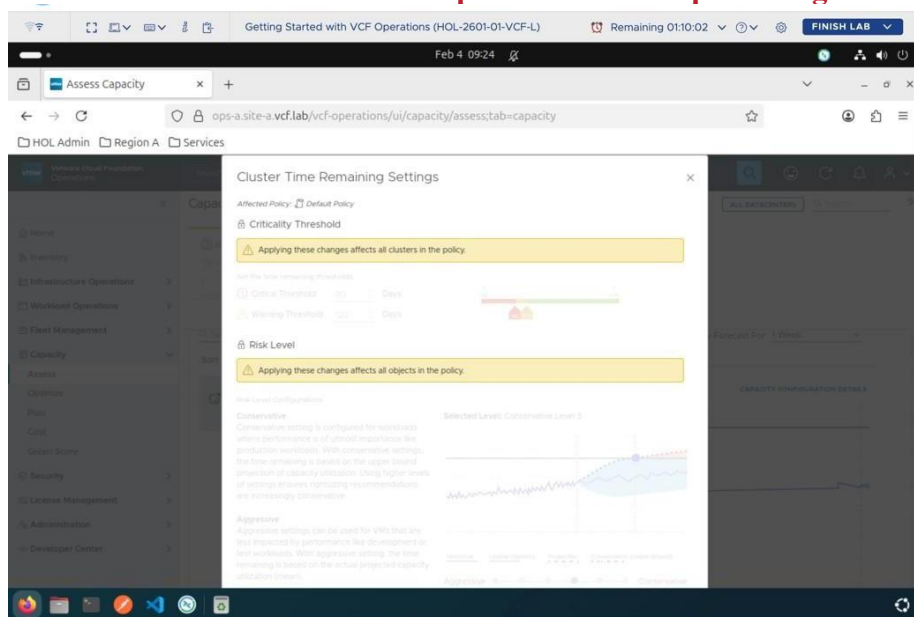
Friday, April 10, 03:55:00 PM

Memory (Demand) Projection Range (MB): 143,005,782.21 - 157,645,624.38
Memory (Demand) Projection (MB): 157,645,624.38
Usable Capacity (MB): 226,489,954

CAPACITY CONFIGURATION DETAILS

1. Change the sort by selection (persons to most Constrained) -- in addition to Most Constrained, we can also sort by CPU Demand, Memory Demand, and Disk Space Demand.
2. This widget shows the capacity data for each vSphere cluster in the datacenter. We can change our selection in this area by selecting another cluster. In the current lab, we have two clusters to choose from.
3. The vertical dotted lines represent today and the day resources will run out (based upon current consumption). In this example, the two lines overlap, as resources will run out today. Everything to the right of the dotted lines represent a trend based on the data Aria Operations has collected previously from the environment.
4. Change the resource selection across the top of the menu (change view between CPU, Memory, and Disk Space). In our screenshot, Memory is alerting as constrained as noted by the "red dot" next to the option while the CPU and Disk Space resources are showing a "green check" noting those resources are currently fine. Note: Depending upon the available resources of the lab, we may see different results than what is shown in this example.
5. Change the Show History For or Show Forecast For options. Available view options include: 1 week, 1 month, 3 months, 6 months, or 12 months. Feel free to choose any timeframe of interest.
6. Click on the CAPACITY CONFIGURATION DETAILS link to see what policy the object is using to determine criticality thresholds.
7. Recommendations (not shown in image) -- scroll further down to see any recommendations Aria Operations makes for the current cluster.
8. RUN SCENARIOS

Hi! How may I help you?



Getting Started with VCF Operations (HOL-2601-01-VCF-L) Remaining 01:10:02 FINISH LAB

Feb 4 09:24

ops-a.site-a.vcf.lab/vcf-operations/ui/capacity/assess;tab=capacity

HOL Admin Region A Services

Cluster Time Remaining Settings

Affected Policy: Default Policy

Criticality Threshold

Applying these changes affects all clusters in the policy.

Set the time remaining thresholds.

Critical Threshold: 10 Days

Warning Threshold: 20 Days

Risk Level

Applying these changes affects all objects in the policy.

High Level Configuration

Conservative: Conservative setting is configured for workload where performance is of utmost importance like production workload. With conservative settings, the time remaining is based on the upper bound projection of capacity utilization. Using higher levels of aggressive settings, the time remaining is based on the actual projected capacity utilization.

Aggressive: Aggressive settings can be used for VMs that are less impacted by performance like development or test workloads. With aggressive settings, the time remaining is based on the actual projected capacity utilization.

Selected Level: Conservative Level 3

Forecast for 1 Week

CAPACITY CONFIGURATION DETAILS

Manual Consoles Resources

cluster in the datacenter. We can change our selection in this area by selecting another cluster. In the current lab, we have two clusters to choose from.

3. The vertical dotted lines represent today and the day resources will run out (based upon current consumption). In this example, the two lines overlap, as resources will run out today. Everything to the right of the dotted lines represent a trend based on the data Aria Operations has collected previously from the environment.

4. Change the resource selection across the top of the menu (change view between CPU, Memory, and Disk Space). In our screenshot, Memory is alerting as constrained as noted by the "red dot" next to the option while the CPU and Disk Space resources are showing a "green check" noting those resources are currently fine. Note: Depending upon the available resources of the lab, we may see different results than what is shown in this example.

5. Change the Show History For or Show Forecast For options. Available view options include: 1 week, 1 month, 3 months, 6 months, or 12 months. Feel free to choose any timeframe of interest.

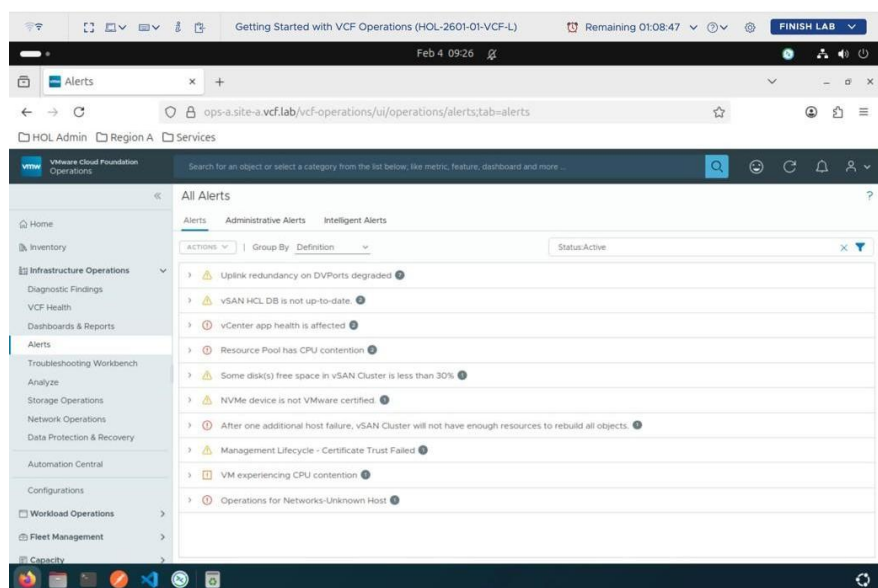
6. Click on the CAPACITY CONFIGURATION DETAILS link to see what policy the object is using to determine criticality thresholds.

7. Recommendations (not shown in image) – scroll further down to see any recommendations Aria Operations makes for the current cluster.

8. RUN SCENARIO (not shown in image) – located at the bottom of this page. This feature is covered in detail in the "Getting More Out of Aria Operations" lab and is outside the scope of this lab's overview.

Hi! How may I help you?

1.15 Alert overview



Getting Started with VCF Operations (HOL-2601-01-VCF-L) Remaining 01:08:47 FINISH LAB

Feb 4 09:28

ops-a.site-a.vcf.lab/vcf-operations/ui/operations/alerts;tab=alerts

HOL Admin Region A Services

All Alerts

Alerts Administrative Alerts Intelligent Alerts

ACTIONS Group By Definition Status: Active

Uplink redundancy on DVPorts degraded

vSAN HCL DB is not up-to-date

vCenter app health is affected

Resource Pool has CPU contention

Some disk(s) free space in vSAN Cluster is less than 30%

NVMe device is not VMware certified

After one additional host failure, vSAN Cluster will not have enough resources to rebuild all objects

Management Lifecycle - Certificate Trust Failed

VM experiencing CPU contention

Operations for Networks-Unknown Host

ALERT OVERVIEW

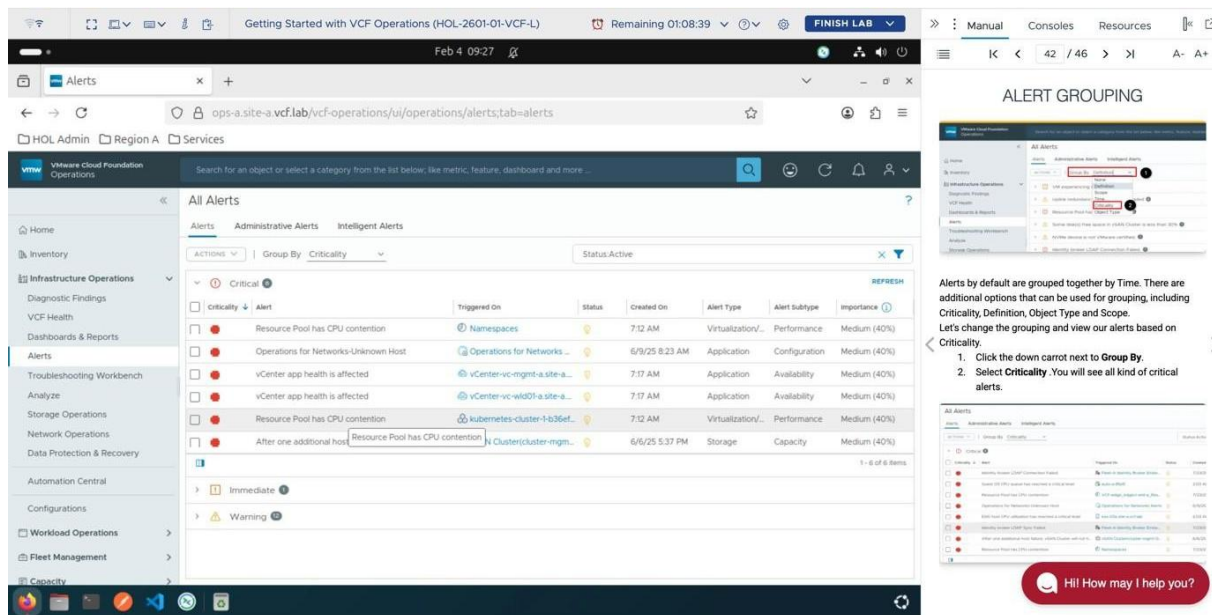
Now, let's locate the Alerts section of the VCF Operations User Interface.

1. Click on Home.
2. Click on the Infrastructure Operations to expand the menu.
3. Click on Alerts.

Hi! How may I help you?

1.16 Alert Grouping

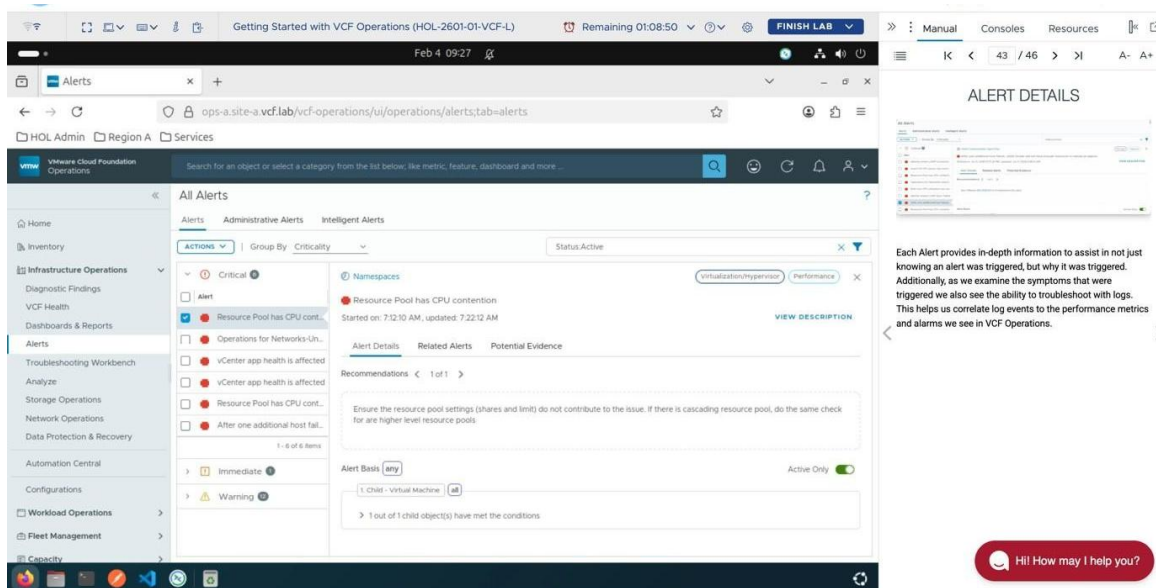
Alerts by default are grouped together by Time. There are additional options that can be used for grouping, including Criticality, Definition, Object Type and Scope.



The screenshot shows the VMware Cloud Foundation Operations console. The left sidebar contains navigation links: Home, Inventory, Infrastructure Operations (Diagnostic Findings, VCF Health, Dashboards & Reports, Alerts, Troubleshooting Workbench, Analyze, Storage Operations, Network Operations, Data Protection & Recovery), Automation Central, Configurations, Workload Operations, Fleet Management, and Capacity. The main area displays 'All Alerts' with a table of alerts. The 'Group By' dropdown is set to 'Criticality'. A right-hand panel titled 'ALERT GROUPING' shows a list of alerts grouped by time. Below this panel, a text box explains that alerts are grouped by time by default and lists two steps to change the grouping: 1. Click the down caret next to Group By. 2. Select Criticality. A chatbot icon at the bottom right says 'Hi! How may I help you?'.

1.17 Alert Details

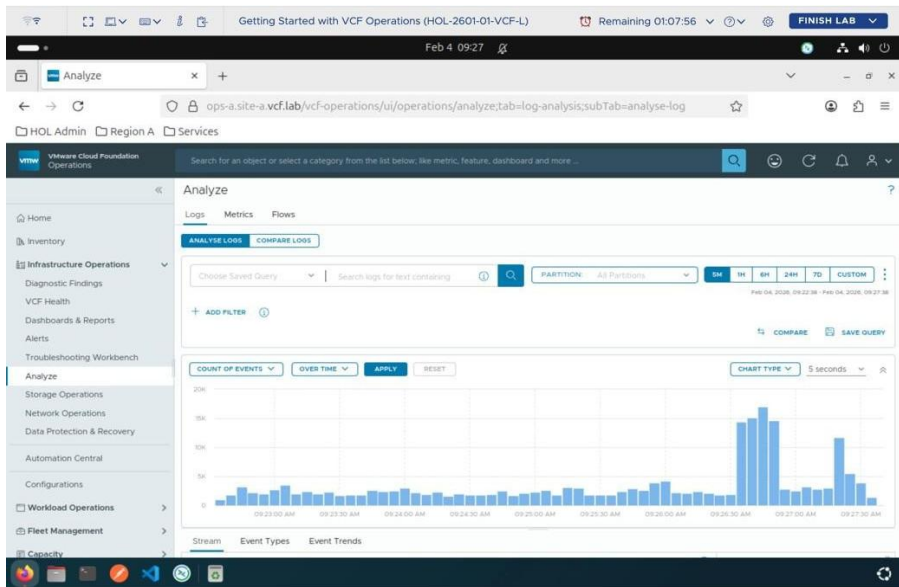
Each Alert provides in-depth information to assist in not just knowing an alert was triggered, but why it was triggered.



The screenshot shows the VMware Cloud Foundation Operations console with the 'Alert Details' view selected. The left sidebar is the same as in the previous screenshot. The main area displays details for a specific alert: 'Resource Pool has CPU contention' in the 'Namespaces' category. It shows the alert was started on 7:12:10 AM and updated on 7:22:12 AM. Below the alert name, there are sections for 'Alert Details', 'Related Alerts', and 'Potential Evidence'. A 'Recommendations' section provides advice on resource pool settings. At the bottom, an 'Alert Basis' section shows the conditions that triggered the alert: '1 Child - Virtual Machine'. A right-hand panel titled 'ALERT DETAILS' shows a log of events related to the alert. A chatbot icon at the bottom right says 'Hi! How may I help you?'.

1.18 Analyze

Many capabilities in VCF Operations become available only when VCF Operations for logs and VCF Operations for networks are deployed and properly configured. One of these capabilities is the new Analyze console



The screenshot shows the VCF Operations Analyze console. The left sidebar contains navigation links for Home, Inventory, Infrastructure Operations, VCF Health, Dashboards & Reports, Alerts, Troubleshooting Workbench, Analyze, Storage Operations, Network Operations, Data Protection & Recovery, Automation Central, Configurations, Workload Operations, Fleet Management, and Capacity. The main area displays the 'Analyze' console with tabs for Logs, Metrics, and Flows. A bar chart shows the 'COUNT OF EVENTS' over time, with a peak around 09:25:30 AM. The right sidebar contains a 'Manual' section with a list of topics: 1. Logs, 2. Metrics, and 3. Flows. A red box highlights the 'Analyze' console title and the 'Logs', 'Metrics', and 'Flows' tabs. A red box also highlights the '1', '2', and '3' numbered list items in the 'Manual' section.

ANALYZE CONSOLE

1. **Logs:** Provides an integrated user experience for log analysis. (Module 2 - Explore and Analyze Logs and build Log Dashboards in VCF Operations)

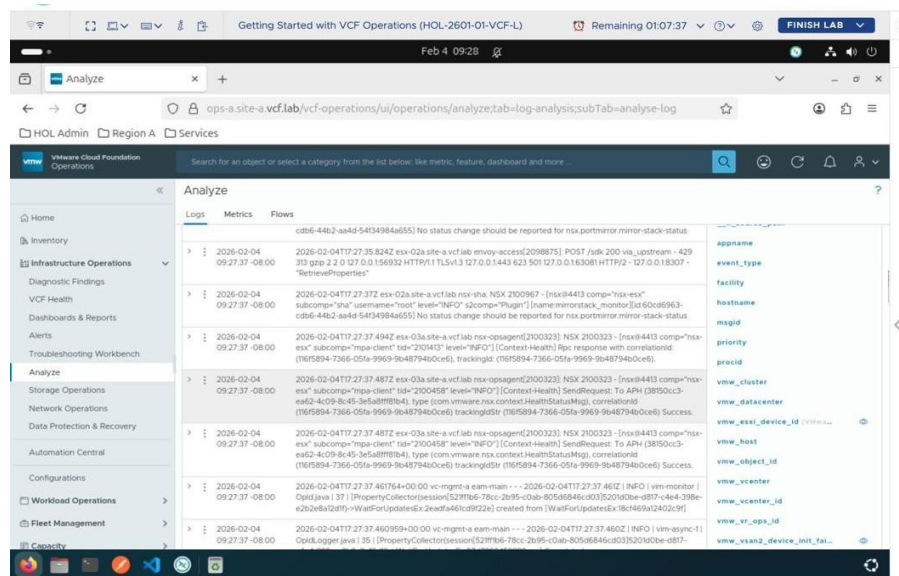
2. **Metrics:** Provides an interface to analyze metrics and objects by creating a query. (Module 3 - Explore and Analyze Object Metrics in VCF Operations)

3. **Flows:** Provides comprehensive visibility into network traffic patterns, performing analysis based on the entities, flows and time range. (Module 4 - Explore and Analyze specific Flows in VCF Operations)

Logs: Provides an integrated user experience for log analysis.

Metrics: Provides an interface to analyze metrics and objects by creating a query.

Flows: Provides comprehensive visibility into network traffic patterns, performing analysis based on the entities, flows and time range.



The screenshot shows the VCF Operations Analyze console with the 'Logs' tab selected. The main area displays a list of log events. The right sidebar contains a 'Manual' section with a list of topics: 1. Logs, 2. Metrics, and 3. Flows. A red box highlights the 'Analyze' console title and the 'Logs', 'Metrics', and 'Flows' tabs. A red box also highlights the '1', '2', and '3' numbered list items in the 'Manual' section.

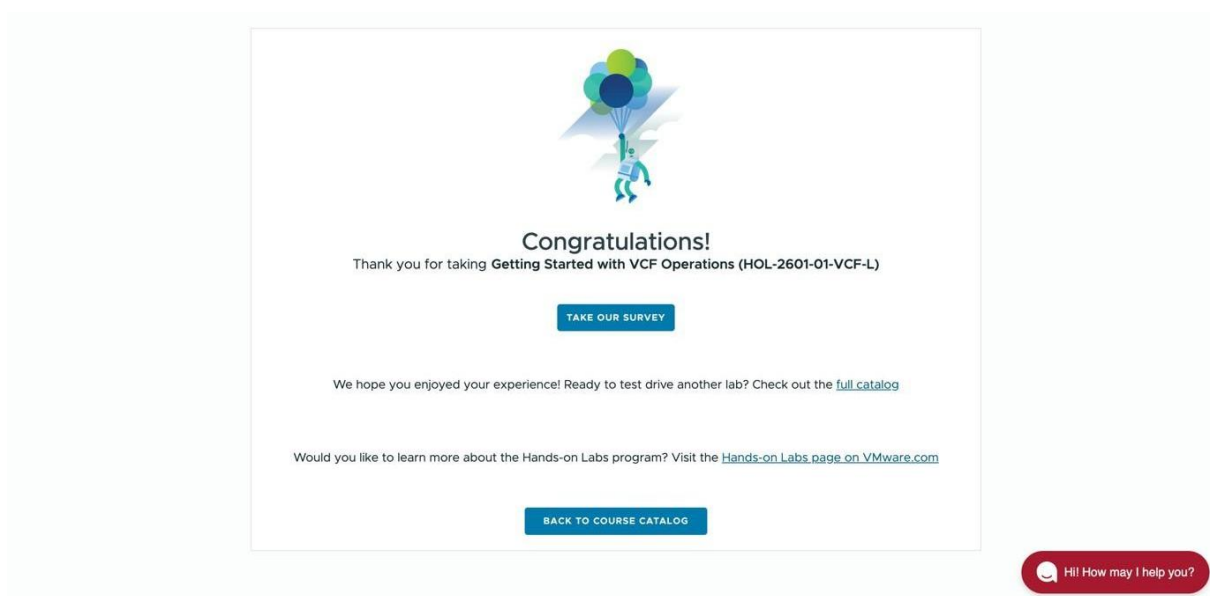
ANALYZE CONSOLE

1. **Logs:** Provides an integrated user experience for log analysis. (Module 2 - Explore and Analyze Logs and build Log Dashboards in VCF Operations)

2. **Metrics:** Provides an interface to analyze metrics and objects by creating a query. (Module 3 - Explore and Analyze Object Metrics in VCF Operations)

3. **Flows:** Provides comprehensive visibility into network traffic patterns, performing analysis based on the entities, flows and time range. (Module 4 - Explore and Analyze specific Flows in VCF Operations)

1.19 Lab Completion



Certificate



VMWARE
HANDS-ON LABS

Certificate of Completion

Shreyans Tatiya

Has Successfully Completed the following Hands-on Lab:

VMware vSAN - Getting Started and Advanced Topics (HOL-2634-01-VCF-L)

Discover what's new in vSAN 9—enable file services, encryption, and monitoring with vCenter and VVCF Operations while managing storage, performance, and Day-2 operations through the intuitive HTML5 interface.

Start Date Time: 06:16 AM 01/29/2026
Completion Date Time: 07:28 AM 01/29/2026
Time Spent: 79 minutes
Language: en



K. J. Somaiya School of Engineering, Mumbai-77
(Somaiya Vidyavihar University)
Department of Computer Engineering



Conclusion:

This lab introduced the core features of VMware vSphere and VMware vCenter Server, covering management, networking, security, and storage.

It provided a solid foundation in virtualization concepts and basic data center operations.

Post Lab Questions:

1. How does VMware Cloud Foundation enhance traditional virtualization?

VMware Cloud Foundation extends traditional virtualization (like VMware vSphere) into a full **Software-Defined Data Center (SDDC)** by integrating:

- Compute (vSphere)
- Storage (VMware vSAN)
- Networking (VMware NSX)
- Automation & lifecycle management

It provides automation, scalability, security, and hybrid cloud support.

2. Why are advanced cloud technologies important for enterprise data centers?

They enable:

- Scalability
- Automation
- Faster deployment
- Stronger security
- Hybrid cloud integration (e.g., Amazon Web Services, Microsoft Azure)

This improves agility, efficiency, and cost control in enterprise data centers.

3. What is the role of DRS and HA in vSphere clusters?

In VMware vSphere:

- **DRS** balances workloads across hosts for optimal performance.
- **HA** restarts VMs automatically if a host fails.

Together, they ensure performance and availability.